ARC-SPRUCE: Post-Quantum Blind Signatures and Anonymous Rate-Limited Credentials from Lattices

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1 INTRODUCTION

Anonymous credentials (AC) are cryptographic protocols that allow parties to convey certified information about themselves in a privacy-preserving way. There are usually three roles in ACs, namely *issuers*, *holders* and *verifiers*. More concretely, an AC scheme allows a holder to prove to a verifier that it possesses certain attributes, which have been authenticated by the issuer. Here, the authenticated attributes are usually called a *credential*, which is kept private by the user. Using a so-called *showing* process, the user can generate a presentation which convinces the verifier that attributes stored in its credential (or credentials) satisfy predicates without revealing any additional information. While there are numerous security properties that ACs can achieve, the most fundamental ones are unforgeability and unlinkability. Here, the former ensures that an issuer indeed generated a credential that a user presents, and that colluding issuers and verifiers may not link a presentation to a specific issuance session. Here, the latter should also ensure that multiple presentations cannot be traced to the same credential.

Certain AC use-cases like rate limiting [YW25, SK25] require that the credential is only valid for a *bounded* number of presentations in a given context (e.g. per epoch or per service) – without revealing which presentations come from the same holder up to the enforced limit. A key advantage of existing, pre-quantum ACs is their performance, e.g. with constant credential and presentation size, regardless of the enforced rate limit. However, the situation looks very different in the post-quantum setting. There, practically efficient AC schemes [BLNS23a, AGJ⁺24, PWFW23], are yet mostly absent. This is highly unfortunate, given the uptake of (privacy-preserving) digital identification schemes in e.g. the EU Digital Identity Wallet, and simultaneous requirement for deployed schemes to eventually be quantum-secure: given current migration timelines, post-quantum ACs may become mandatory in the next 5 years or fewer. Recently, Google [AS26] moved the timeline for post-quantum to 2029. While everlasting anonymous rate-limited tokens [CADLT25] provide post-quantum privacy, they only guarantee rate limiting until quantum computers arrive.

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Building ACs. To understand how post-quantum ACs (with rate-limiting) are usually constructed, it is instructive to understand how classically-secure constructions work. Anonymous credentials are often built on top of a Blind Signature design. Here, one of the most well-known approach (which is also the most promising in the post-quantum setting) is due to Fischlin [Fis06]:

(1) The issuer initially creates a key pair $(\mathsf{sk}, \mathsf{vk})$ using an EUF CMA-secure signature scheme. We assume that user and verifier have vk, while the issuer keeps sk.
(2) The holder first commits to its message μ as t, using randomness r. It then sends t to the issuer.
(3) Upon receiving t, the issuer signs it and thus creates a signature sig. It then sends sig to the user, who checks its verifiability with vk.
(4) To create the blind signature, the user creates a Non-Interactive Zero-Knowledge Proof of Knowledge (NIZK) showing that it knows sig, r such that

$$\mathsf{Verify}_{\mathsf{vk}}(t, \mathsf{sig}) = 1 \land t = \mathsf{Commit}(\mu; r)$$

Combining existing post-quantum primitives generically, even with zero-knowledge friendly hash functions [PWFW23], leads to non-competitive performance. Thus, existing works carefully choose commitment schemes, signatures and NIZKs such that the NIZK for the given relation has good proof size and prover/verifier time. However, the state-of-the-art is generally not considered efficient enough for practical deployments: Constructing blind signatures from alternative post-quantum hash-and-sign cryptography [BFM⁺25] and anonymous credentials from lattices [LCL⁺23], where recent work brings the credential size down to just over 70 kB using optimized samplers [JRS23, JS25].

Our construction builds on proof-friendly lattice signatures from vanishing SIS (vSIS) with rational hashing [DKLW25a, DKLW25b] and on fast trapdoor generation and Gaussian preimage sampling for NTRU-like lattices [ENS⁺23]. For succinct zero knowledge over small fields, we use the SmallWood framework for hash-based polynomial commitments and compiled PIOPs [FR25b]. While SmallWood has been used for ZK-friendly post-quantum signatures in CAPSS [FR25a], we use a zk-friendly PRF with a significantly smaller modulus for tag derivation, which reduces the size of our communication significantly, and discuss Poseidon/Poseidon2-style instantiations [GKR⁺21, GKS23].

1.1 Contributions

In this work, we narrow the efficiency gap for post-quantum blind signatures and ACs substantially. For Blind Signatures, we achieve this by optimizing existing design ideas carefully and employ novel primitives. We then extend our construction to yield the first post-quantum rate-limiting anonymous credentials. Crucially, the size of our credential as well as the presentation is essentially independent of the rate limit.

We construct a Blind Signature based on a signature whose security relies on lattice assumptions together with a hash-based

NIZK. We then extend this to an *Anonymous Rate-Limited Credential* (ARC) [YW25] by including a Pseudorandom Function (PRF) into our design. Using a key committed in the message that is signed by the issuer, the user is able to evaluate a PRF on the key on different nonces. This PRF output, which we call a *tag*, can be made public. Since the signature and thus the key remain hidden due to the use of a NIZK, this intuitively makes the scheme unlinkable. By including a proof of correct PRF evaluation into the NIZK, the user also cannot generate a differing tag for the same nonce. Therefore, any two presentations generated for the same credential are linkable as they must have the same tag, which makes the scheme rate-limiting.

To design the Blind Signature, we follow the recent proposal of [BLNS23b, LSS24, DKLW25b] which also presents blind signatures and anonymous credentials based on lattice assumptions. We first let the holder commit to its message μ using an Ajtai commitment. Then, in an interactive process, issuer and signer will sample randomness which together with the committed message becomes a target $\mathbf{t}$ that will be signed by the issuer. Here, the user proves in zero knowledge that $\mathbf{t}$ is consistent with the committed message and the issuer-supplied randomness. The signature scheme is formed by a matrix $\mathbf{A}$ together with a trapdoor td that allows sampling of short preimages $\bmod q$ using a GPV-style [GPV08] preimage sampler. The issuer then uses td to sample a short preimage $\mathbf{u}$ such that $\mathbf{Au} = \mathbf{t} \bmod q$, which serves as the signature. To prove knowledge of a valid $\mathbf{u}$ we then use the SmallWood NIZK scheme [FR25b]. While this approach makes the round complexity of issuance non-optimal (requiring 3 rounds instead of 1 round), it allows tighter parameters for the preimage sampler (as $\mathbf{t}$ is uniform) while making the statement proven in the NIZK structurally simple. To turn this construction into an ARC, we let the user additionally commit to a PRF key $\mathbf{k}$ for the Poseidon2 hash function [GKS23]. It turns out that the parameters such as the modulus q of Poseidon2 can be chosen to be compatible with those of the lattice commitment, signature scheme and NIZK, making the proof of PRF evaluation highly efficient. The holder can then attach an evaluation of the PRF on a nonce where the committed key is used, together with a proof of correct PRF evaluation, as part of the presentation. If the holder reuses the nonce, this will be noticed by a verifier keeping track of previously issued tags.

1.2 Technical overview

This work constructs ARC around a single protocol blueprint with message $\mathbf{m} = \mu \| \mathbf{k}$, where μ is a vector of attributes and $\mathbf{k}$ is a *secret* PRF key.

Issuance. Let $\mathbf{A}_{\mathsf{Com}}$ be a uniformly random and compressing matrix to be used for the Ajtai commitment. In the issuance protocol, the holder additionally samples short $\mathbf{r}_0^{\mathcal{H}}, \mathbf{r}_1^{\mathcal{H}}, \mathbf{r}$ to compute the commitment $\mathbf{c} = \mathbf{A}_{\mathsf{Com}}[\mu \| \mathbf{r}]$ that is sent to the issuer. It, in response, samples short $\mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}}$ that are returned to the holder. The target $\mathbf{t}$ is then computed as

$$\mathbf{t} := hash(\mu, \mathbf{k}, \mathsf{center}\mathbf{r}_0^{\mathcal{H}} + \mathbf{r}_0^{\mathcal{I}}, \mathsf{center}\mathbf{r}_1^{\mathcal{H}} + \mathbf{r}_1^{\mathcal{I}})$$

where center derives a short value from the sum and *hash* is a so-called rational hash function as introduced in [BLNS23b]. The holder then sends $\mathbf{t}$ to the issuer, together with a NIZK that it is well-formed according to $\mathbf{c}, \mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}}$ and that all involved values are short.

The signature scheme is based on a hash-and sign signature. It precomputes preimages of a matrix $\mathbf{A}$ using a trapdoor Gaussian preimage sampler over an NTRU-like lattice (Antrag/hybrid sampling) to instantiate the issuer's trapdoor signing step. The short preimage $\mathbf{u}$ can then be returned to the holder, who verifies that $\mathbf{t} = \mathbf{Au}$.

We illustrate the issuance transcript flow and pre-sign proofs in Figure 1.

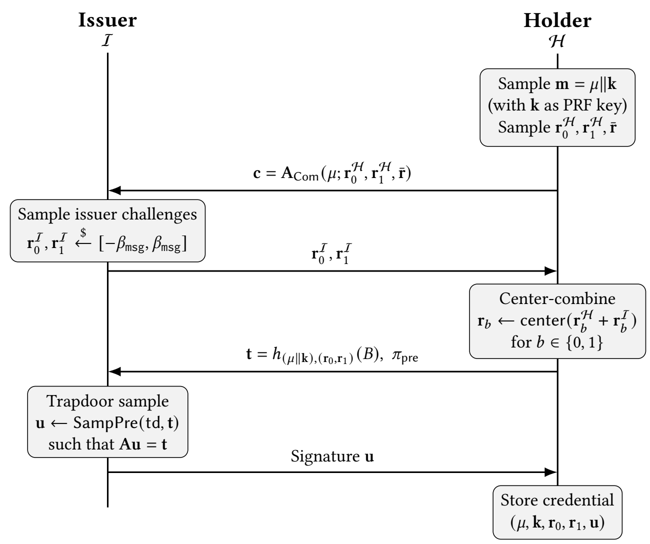

Figure 1: Issuance flow (blind signature). The pre-sign proof π_{pre} binds t to the commitment and the centered randomness, while hiding μ.

Blind Signatures and ARC Presentations. Let $\mathbf{r}_0 := \mathsf{center}(\mathbf{r}_0^{\mathcal{H}} + \mathbf{r}_0^{\mathcal{I}}), \mathbf{r}_1 := \mathsf{center}(\mathbf{r}_1^{\mathcal{H}} + \mathbf{r}_1^{\mathcal{I}})$. A presentation is a NIZK showing knowledge of $\mathbf{m}, \mathbf{r}_0, \mathbf{r}_1, \mathbf{t}$, such that $\mathbf{t} = hash(\mathbf{m}, \mathbf{r}_0, \mathbf{r}_1)$, as well as $\mathbf{u}$ such that $\mathbf{t} = \mathbf{Au}$ where $\mathbf{m}, \mathbf{r}_0, \mathbf{r}_1, \mathbf{u}$ are short. Towards constructing ARC, we additionally compute $\mathsf{tag} = \mathsf{F}(\mathbf{k}; \mathsf{nonce})$, append it to the NIZK and additionally prove in the NIZK that nonce is in the rate limiting interval and that tag is well-formed.

The verifier checks the NIZK proof and enforces the rate policy by maintaining server-side state keyed by $(\mathsf{nonce}, \mathsf{tag})$ (or an application-specific encoding thereof) to avoid double-spending.

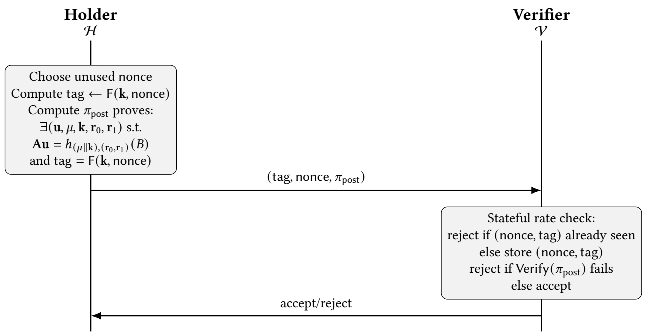

Figure 2: Showing (ARC presentation). Using an unused $\mathsf{nonce} \leq \mathsf{presentation_{limit}}$**, the holder computes a deterministic** tag **from** $(\mathbf{k}, \mathsf{nonce})$ **(rate limiting) and pseudorandom across unused nonces (unlinkability).**

Figure 2 summarizes how tags enable stateful rate limiting without revealing the credential.

SmallWood Proofs. SmallWood [FR25b] is a hash-based polynomial commitment scheme which produces efficient witnesses for polynomials of small degree $\leq 2^{16}$, which is particularly useful for lattice-based problems. The construction avoids the overhead of asymptotically-designed systems like STARKS while achieving better concrete efficiency than MPC-in-the-head variants for small witnesses. SmallWood exclusively relies on hash-based primitives, making it plausibly post-quantum secure. We use SmallWood as an NIZK for polynomial constraints over a small field $\mathbb{F}_q$, and we treat it as a black box that provides: (i) commitment to witness rows packed on an evaluation domain, (ii) succinct openings at verifier-chosen points, and (iii) soundness and zero knowledge after Fiat–Shamir [FR25b].

To obtain compact proofs, both issuance and showing are compiled into SmallWood statements over committed witness rows, with explicit bridge families for the linear commitment, centering, rational-hash relation, signature relation, and PRF evaluation. The exact compiled relations are specified in Section 5, while proof-system internals are deferred to Appendix C.

In the showing proof, the rational-hash/signature relation is realized through a full transform-basis replay image whose vanishing implies the corresponding coefficient-domain cleared relation by the theorem stated in Section 5.

2 PRELIMINARIES

This section defines notation and gives interface-level definitions for the lattice primitives, the NIZK model, the blind-signature syntax, ARC syntax and security goals, and pseudorandom functions.

2.1 Notation

Let λ denote the security parameter and $\mathrm{negl}(\lambda)$ be any positive function negligible in λ. We write PPT for probabilistic algorithms whose runtime is polynomial in λ. For integers $a < b$ we write $[a, b] \coloneqq \{a, a+1, \ldots, b\}$ and we use $[b]$ as shorthand for $[1, b]$. Throughout this work, we denote vectors in lower-case bold letters such as $\mathbf{b}$ and matrices in upper-case bold letters such as $\mathbf{A}$.

We work over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ for power-of-two N and its quotient $R_q = R/qR$, identified with polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$. For $a \in \mathbb{Z}$, we write $\langle a \rangle_q \in \{0, \ldots, q-1\}$ for reduction modulo q. When considering $\mathbb{Z}_q$-elements from an interval $[-a, b]$, we identify these with the set $\{q-a, q-a+1, \ldots, 0, 1, \ldots, b\}$. For elements $x \in R_q$ we write $||x|| \leq B$ to mean that each coefficient of x comes from $[-B, B]$. For vectors $\mathbf{x} \in R_q^\ell$, $||\mathbf{x}|| \leq B$ means that the statement holds for all elements of $\mathbf{x}$.

2.2 Commitment schemes and Ajtai commitments

Definition 2.1 (Commitment scheme)**.** Let $\mathcal{M}$ be a finite message space. A (non-interactive) commitment scheme is a tuple of PPT algorithms

$$\mathsf{Com} = (\mathsf{Setup}, \mathsf{Commit}, \mathsf{Verify})$$

with the following syntax:

- $\mathsf{pp}_{\mathsf{Com}} \leftarrow \mathsf{Setup}(1^\lambda)$ is a PPT algorithm which outputs public parameters $\mathsf{pp}_{\mathsf{Com}}$. It is implicit input to all other algorithms.
- $c \leftarrow \mathsf{Commit}(m)$ is a PPT algorithm which takes a message $m \in \mathcal{M}$ as input and outputs a commitment c and decommitment d.
- $\mathsf{Verify}(c, m, d) \in \{0, 1\}$ deterministically checks validity of the opening.

Correctness holds if for all $\mathsf{pp}_{\mathsf{Com}} \leftarrow \mathsf{Setup}$, $m \in \mathcal{M}$ and $(c, d) \leftarrow \mathsf{Commit}(m)$ it holds that

$$\mathsf{Verify}(c, m, d) = 1.$$

The scheme is *computationally binding* if for all PPT adversaries $\mathcal{A}$,

$$\Pr_r\left[\begin{matrix}(\mathsf{pp}_{\mathsf{Com}}, c, m, d, m', d') \leftarrow \mathcal{A}(r) \wedge m \neq m' \\ \mathsf{Verify}(c, m, d) = 1 \wedge \mathsf{Verify}(c, m', d') = 1\end{matrix}\right] \leq \mathrm{negl}(\lambda).$$

The scheme is *(computationally) hiding* if for all PPT distinguishers $\mathcal{D}$, all messages $m_0, m_1 \in \mathcal{M}$ and $\mathsf{pp}_{\mathsf{Com}} \leftarrow \mathsf{Setup}()$,

$$\left|\Pr[\mathcal{D}^{O_{\mathsf{Commit}}(m_0)}(\mathsf{pp}_{\mathsf{Com}}) = 1] - \Pr[\mathcal{D}^{O_{\mathsf{Commit}}(m_1)}(\mathsf{pp}_{\mathsf{Com}}) = 1]\right| \leq \mathrm{negl}(\lambda)$$

where $O_{\mathsf{Commit}}(m_b)$ runs $(c, d) \leftarrow \mathsf{Commit}(m_b)$ and outputs c and the probability is taken over the randomness of Commit, $\mathcal{D}$.

2.2.1 Ajtai’s Commitment scheme. We use the standard bounded linear commitment over R_q, matching the compact Ajtai-style commitment of [ACK21, Definition 13]. Fix message length ℓ_m, randomness dimension ℓ_r, output length ℓ_{Com}, and coefficient bound β_{msg}, and let $\ell_{\mathsf{in}} = \ell_m + \ell_r$.

- $\mathsf{Setup}(1^\lambda)$ samples $\mathbf{A}_{\mathsf{Com}} \xleftarrow{\$} R_q^{\ell_{\mathsf{Com}} \times \ell_{\mathsf{in}}}$ and outputs $\mathsf{pp}_{\mathsf{Com}} \coloneqq (\mathbf{A}_{\mathsf{Com}}, \beta_{\mathsf{msg}})$.
- The message space is

$$\mathcal{M} \coloneqq \{\mathbf{m} \in R_q^{\ell_m} \mid \|\mathbf{m}\|_\infty \leq \beta_{\mathsf{msg}}\}.$$

- $\mathsf{Commit}(\mathbf{m})$ samples $\mathbf{r} \xleftarrow{\$} [-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]^{\ell_r} \subseteq R_q^{\ell_r}$ and outputs

$$\mathbf{c} = \mathbf{A}_{\mathsf{Com}} \cdot [\mathbf{m} \| \mathbf{r}]^\top$$

together with opening $\mathbf{r}$.

- Verify$(\mathbf{c}, \mathbf{m}, \mathbf{r})$ accepts iff
$$\mathbf{c} \in R_q^{\ell_{\mathsf{Com}}}, \qquad \mathbf{m} \in \mathcal{M}, \qquad \|\mathbf{r}\|_\infty \le \beta_{\mathtt{msg}},$$
and $\mathbf{c} = \mathbf{A}_{\mathsf{Com}} \cdot [\mathbf{m}\|\mathbf{r}]^\top$.

Definition 2.2 (Module-SIS and module-ISIS over R_q)**.** Fix integers $n, m \ge 1$, modulus $q \ge 2$, and bound $\beta > 0$.

- $\mathsf{MSIS}_\infty(n, m, q, \beta)$. Given $\mathbf{A} \in R_q^{n\times m}$, find a non-zero $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} \equiv 0 \pmod q$ and $\|\mathbf{s}\|_\infty \le \beta$.
- $\mathsf{MISIS}_\infty(n, m, q, \beta)$. Given $(\mathbf{A}, \mathbf{t}) \in R_q^{n\times m} \times R_q^n$, find $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} \equiv \mathbf{t} \pmod q$ and $\|\mathbf{s}\|_\infty \le \beta$.

Lemma 2.3 (Binding of the Ajtai commitment)**.** *Suppose an adversary outputs a commitment* $\mathbf{c}$ *and two distinct valid openings* $(\mathbf{m}, \mathbf{r}) \ne (\mathbf{m}', \mathbf{r}')$ *with coefficient bound* $\beta_{\mathtt{msg}}$*. Then*
$$\Delta := [\mathbf{m} - \mathbf{m}'\|\mathbf{r} - \mathbf{r}']^\top$$
is a non-zero solution to MSIS_∞ *for* $\mathbf{A}_{\mathsf{Com}}$ *with bound* $2\beta_{\mathtt{msg}}$*, since* $\mathbf{A}_{\mathsf{Com}}\Delta \equiv 0 \pmod q$ *and* $\|\Delta\|_\infty \le 2\beta_{\mathtt{msg}}$*. Hence the commitment is computationally binding whenever* MSIS_∞ *is hard for the resulting parameters [ACK21, Lemma 2].*

Lemma 2.4 (Statistical hiding of the Ajtai commitment)**.** *Let* B_r *denote the coefficient bound of the commitment-randomness alphabet, so that the commitment randomness is sampled coefficient-wise from* $[-B_r, B_r]$*. If*
$$N\ell_r \log_2(2B_r + 1) \;\ge\; N\ell_{\mathsf{Com}} \log_2 q + 2\lambda,$$
then the commitment distribution is statistically $2^{-\lambda}$*-close to a distribution independent of the committed message. In the present instantiation the same coefficient bound is used for messages and commitment randomness, so* $B_r = \beta_{\mathtt{msg}}$*, and a sufficient condition is*
$$N\ell_r \log_2(2\beta_{\mathtt{msg}} + 1) \;\ge\; N\ell_{\mathsf{Com}} \log_2 q + 2\lambda.$$

Proof. This is the direct specialization of [ACK21, Lemma 1] to the present notation. The randomness block contains $N\ell_r$ coefficients, each uniformly distributed in an alphabet of size $2B_r + 1$, hence its min-entropy is $N\ell_r \log_2(2B_r + 1)$. The commitment output lies in $R_q^{\ell_{\mathsf{Com}}}$, whose support size is $q^{N\ell_{\mathsf{Com}}}$, so the output entropy term is $N\ell_{\mathsf{Com}} \log_2 q$. The leftover-hash loss is 2λ bits, which yields the displayed condition. □

2.3 Lattices, trapdoors, vSIS, and rational hashing

We recall only the definitions needed to state our construction; implementation details of the trapdoor sampler are deferred to Appendix B.

Definition 2.5 (Lattice trapdoor suite)**.** A lattice trapdoor suite over R_q consists of PPT algorithms
$$(\mathsf{TrapGen}, \mathsf{SampPre})$$
and admissible public parameters $(n, m, q, \beta_{\mathtt{sig}})$ as in [DKLW25a, Definition 1] such that:

- $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda)$ outputs a public matrix $\mathbf{A} \in R_q^{n\times m}$ and a trapdoor td, where the distribution of $\mathbf{A}$ is statistically close to uniform over $R_q^{n\times m}$.
- On input $(\mathsf{td}, \mathbf{t})$ with $\mathbf{t} \in R_q^n$, the algorithm $\mathsf{SampPre}(\mathsf{td}, \mathbf{t})$ outputs $\mathbf{u} \in R^m$ such that $\mathbf{A}\mathbf{u} = \mathbf{t}$ in R_q^n, except with negligible failure probability, and $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible probability [GPV08, DKLW25a].

Definition 2.6 (Trapdoor rational hash)**.** A trapdoor rational hash scheme over R_q is a tuple
$$\mathsf{TDRH} = (\mathsf{Setup}, \mathsf{TrapGen}, \mathsf{Eval}, \mathsf{SampPre})$$
consisting of the lattice trapdoor suite above together with a public evaluation algorithm. On input 1^λ, Setup outputs a public hash description B, a message space $\mathcal{M}$, a randomness space $\mathcal{X}$, an admissibility error bound δ, and public bounds $(\beta_{\mathtt{msg}}, \beta_{\mathtt{sig}})$. For fixed B, the algorithm $\mathsf{Eval}(B, m, \chi)$ takes a message $m \in \mathcal{M}$ and witness randomness $\chi \in \mathcal{X}$ and outputs a target $\mathbf{t} \in R_q^n$ or $\perp$ if the input is inadmissible.

For a public key B, define the admissible message subspace
$$\mathcal{M}_B := \left\{ m \in \mathcal{M} : \Pr_{\chi \leftarrow \mathcal{X}} \left[\mathsf{Eval}(B, m, \chi) = \perp\right] \le \delta \right\},$$
following the Σ_{vSIS} template of [DKLW25a, Section 4.1]. Correctness requires that for every $m \in \mathcal{M}_B$ and every $\chi \in \mathcal{X}$ with $\mathsf{Eval}(B, m, \chi) = \mathbf{t} \ne \perp$, the output $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, \mathbf{t})$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{t}$ and $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$, except with negligible probability.

Definition 2.7 (ARC rational hash, admissibility, and cleared form)**.** Our concrete instantiation uses a public key $B = (B_0, B_1, B_2, B_3) \in R_q^4$, message $m \in R_q$, and witness randomness $(r_0, r_1) \in R_q^2$ with coefficient bounds $\|m\|_\infty, \|r_0\|_\infty, \|r_1\|_\infty \le \beta_{\mathtt{msg}}$. Write
$$\mathrm{num}(B, m, r_0) := B_0 + B_1 m + B_2 r_0, \qquad \mathrm{den}(B, r_1) := B_3 - r_1.$$
We say that (B, m, r_0, r_1) is *admissible* if $\mathrm{den}(B, r_1) \in R_q^\times$, equivalently, if $\mathrm{den}(B, r_1)$ is non-zero in every CRT slot of R_q.

When this admissibility condition holds, the rational hash is the partial function
$$h_{m,(r_0,r_1)}(B) := \mathrm{num}(B, m, r_0) \oslash \mathrm{den}(B, r_1).$$

For an arbitrary target $T \in R_q$, we also define the cleared-denominator relation
$$(B_3 - r_1) \odot T - \big(B_0 + B_1 m + B_2 r_0\big) = 0. \tag{1}$$
The cleared relation is a well-formed ring identity for all inputs. It is used below as a compiled proof-system relation.

Lemma 2.8 (Cleared form under admissibility)**.** *Let*
$$R_q \cong \prod_{i=1}^{s} F_i$$
be the CRT decomposition of R_q *into field factors, and write*
$$\mathrm{num}(B, m, r_0) = (n_1, \ldots, n_s), \qquad \mathrm{den}(B, r_1) = (d_1, \ldots, d_s), \qquad T = (t_1, \ldots, t_s).$$
If (B, m, r_0, r_1) *is admissible, that is,* $d_i \ne 0$ *for every* i*, then*
$$T = h_{m,(r_0,r_1)}(B) \iff (B_3 - r_1) \odot T - \big(B_0 + B_1 m + B_2 r_0\big) = 0.$$
Without admissibility, the implication from right to left may fail.

Proof. Under the CRT isomorphism, the defined quotient relation is the coordinate-wise statement
$$t_i = n_i / d_i \qquad \text{for every } i.$$

If $d_i \neq 0$ for every i, then each d_i is invertible in F_i, so

$$t_i = n_i/d_i \iff d_i t_i - n_i = 0.$$

Collecting the coordinates yields the equivalence with the cleared identity. If admissibility fails, then $d_i = 0$ in at least one slot. In that slot the cleared equation reduces to $0 \cdot t_i - n_i = 0$, hence to $n_i = 0$, and imposes no condition on t_i. The quotient remains undefined there, so the cleared relation is no longer sufficient. □

Remark 2.9 (Why the admissibility side condition is necessary). The cleared identity is therefore a necessary condition for the defined quotient relation, but it is not a sufficient condition in general. A concrete counterexample is obtained by choosing a CRT slot i for which $d_i = 0$, $n_i = 0$, and t_i arbitrary. Then the cleared identity holds in slot i, but the quotient n_i/d_i is undefined. All later uses of the rational hash therefore state explicitly whether they refer to the abstract defined quotient relation or to the weaker cleared relation.

Remark 2.10 (Relation to the DKLW25 BBS candidate). Our ARC family is exactly the BBS-type rational hash studied in [DKLW25a, Table 1], specialized by

$$\tilde{b}_0 = (B_0, B_1, B_2), \qquad \tilde{b}_1 = B_3,$$
$$u = m, \qquad x_0 = r_0, \qquad x_1 = r_1.$$

Equivalently,

$$h_{m,(r_0,r_1)}(B) = \frac{B_0 + B_1 m + B_2 r_0}{B_3 - r_1}.$$

The undefined case $h_{m,(r_0,r_1)}(B) = \perp$ is exactly the inadmissibility event that $B_3 - r_1$ is not invertible slotwise, and is absorbed by the admissibility parameter δ. In issuance, (r_0, r_1) is obtained from the holder and issuer contributions via the centering map of Appendix A; by Lemma A.1, this is compatible with the bounded randomness space used in the DKLW security experiments.

Definition 2.11 (vSIS-style hash-and-preimage signature template)**.** Fix a trapdoor rational hash scheme TDRH with public output $(B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\mathtt{msg}}, \beta_{\mathtt{sig}})$. The associated signature scheme vSIS $=$ (KG, Sign, Verify) is the Σ_{vSIS} template of [DKLW25a, Figure 4 and Section 4.1], specialized to the public function description B and the evaluation map Eval:

- $\mathsf{KG}(1^\lambda)$ runs $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda)$ and outputs $\mathsf{vk} = \mathbf{A}$, $\mathsf{sk} = \mathsf{td}$.
- $\mathsf{Sign}(\mathsf{sk}, m)$ for $m \in \mathcal{M}_B$ samples $\chi \leftarrow \mathcal{X}$ until $\mathbf{t} = \mathsf{Eval}(B, m, \chi) \neq \perp$, computes $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, \mathbf{t})$, and outputs $\sigma = (\mathbf{u}, \chi)$.
- $\mathsf{Verify}(\mathsf{vk}, m, \sigma)$ for $\sigma = (\mathbf{u}, \chi)$ accepts iff $m \in \mathcal{M}_B$, $\chi \in \mathcal{X}$, $\mathsf{Eval}(B, m, \chi) \neq \perp$, $\|\mathbf{u}\|_\infty \leq \beta_{\mathtt{sig}}$, and
$$\mathbf{A}\mathbf{u} = \mathsf{Eval}(B, m, \chi).$$

Definition 2.12 (Hinted-vSIS and GenISIS$_f$ variants from DKLW25)**.** We recall the DKLW25 assumption variants used in the reduction discussion below. Fix a public left factor F, hint space H, target space G, predicate P, and query bound Q, all as in [DKLW25a, Definition 5].

- **Hint-vSIS.** The adversary chooses a query set $\boldsymbol{Q} \subseteq H$ and a fresh target function $g^\star \in G \setminus \boldsymbol{Q}$ before seeing the sampled matrix A. It then receives short preimages of the queried hints $q(A)$ under $F(A)$ for every $q \in \boldsymbol{Q}$, and must output a non-zero short vector $u^\star$ such that $F(A)u^\star = g^\star(A)$. The instance is restricted by the predicate condition $P(F \cup \boldsymbol{Q} \cup (\{g^\star\} \setminus \{0\})) = 1$.
- **s-Hint-vSIS.** This is the strong variant of the same experiment. The adversary still chooses the hint set $\boldsymbol{Q}$ before seeing A, but it is allowed to choose the fresh target $g^\star$ only after seeing A and the hinted preimages.
- **s-\$-Hint-vSIS.** This is the strong random-hinted variant. It is the same as s-Hint-vSIS, except that the query set $\boldsymbol{Q}$ is sampled uniformly at random from H rather than chosen by the adversary [DKLW25a, Definition 5].

Now fix a keyed evaluation family $f : K \times M \times X \to R_q^n$ as in [DKLW25a, Definition 8].

- **GenISIS$_f$.** The challenger samples a public key $\kappa \leftarrow K$ and a uniform matrix $A \xleftarrow{\$} R_q^{n \times m}$. It provides a preimage oracle that, on input $\mu \in M$, samples $\chi \leftarrow X$, sets $t = f(\kappa, \mu, \chi)$, returns a short preimage s such that $As = t$, and records the tuple (s, μ, χ). The adversary wins if it outputs a fresh tuple $(s^\star, \mu^\star, \chi^\star)$ not previously returned by the oracle such that $As^\star = f(\kappa, \mu^\star, \chi^\star)$ and $0 < \|s^\star\| \leq \beta$.
- **IntGenISIS$_f$.** DKLW25 also defines an interactive extension in which the challenger offers a signing-style oracle with an additional linear masking term, and Theorem 6 shows how this interactive problem reduces back to GenISIS$_f$ under enlarged parameters.

Definition 2.13 (Non-blind signature experiments for Σ_{vSIS})**.** We use the signature-security experiments of [DKLW25a, Definition 4 and Figure 3] specialized to the template of Definition 2.11. The load-bearing experiment for the present paper is strong existential unforgeability under random-message attack.

Fix $Q \in \mathrm{poly}(\lambda)$. In the experiment $\mathsf{Exp}^{\Sigma_{\mathrm{vSIS}},\mathcal{A},Q}_{\mathrm{sEUF\text{-}RMA}}(\lambda)$, the challenger samples the public parameters and a key pair, then draws $m_1, \ldots, m_Q$ independently and uniformly from the admissible message space $\mathcal{M}_B$. For each $i \in [Q]$ it returns a signature $\sigma_i \leftarrow \mathsf{Sign}(sk, m_i)$. Finally $\mathcal{A}$ outputs a pair $(m^\star, \sigma^\star)$ and wins iff:

(1) $\mathsf{Verify}(vk, m^\star, \sigma^\star) = 1$, and
(2) $(m^\star, \sigma^\star) \notin \{(m_i, \sigma_i) : 1 \leq i \leq Q\}$.

The experiments sEUF-CMA, sEUF-SMA, and sSUF-SMA are exactly those of [DKLW25a, Figure 3].

Lemma 2.14 (Theorem-backed status of the non-blind rational-hash signature layer)**.** *Let* $\ell_{\mathrm{hash}} = 4$ *for the present ARC instantiation and set*

$$k_{\mathrm{DKLW}} := m + \ell_{\mathrm{hash}}.$$

Assume the following:

(1) $(n, m, q, \beta_{\mathtt{sig}})$ *are admissible parameters for* (TrapGen, SampPre)*;*
(2) the inadmissibility bound satisfies $\delta \leq \mathrm{negl}(\lambda)$*;*
(3) a DKLW-compatible norm bound β_{DKLW} *is fixed so that every signature accepted by the present verifier also satisfies* $\|\mathbf{u}\| \leq \beta_{\mathrm{DKLW}}$ *in the norm used in [DKLW25a];*
(4) the predicate condition of [DKLW25a, Definition 7] holds for the instantiated rational-hash family; and
(5) the denominator-bound and strong-linear-independence side conditions required by [DKLW25a, Theorems 4–8 and Corollary 1] hold for the instantiated family.

Then the following theorem-backed claims hold.

(1) *The non-blind template of Definition 2.11 is correct by [DKLW25a, Theorem 4].*
(2) *Under the strong random-hinted vSIS assumption of [DKLW25a] for the induced parameter tuple, the non-blind template is* sEUF-RMA *by [DKLW25a, Theorem 5].*
(3) *For* $f(B, m, \chi) = h_{m,(r_0,r_1)}(B)$*, the experiment* $\mathsf{GenISIS}_f$ *is equivalent to the non-blind* sEUF-RMA *experiment by [DKLW25a, Theorem 7].*
(4) $\mathsf{GenISIS}_f$ *implies the same strong random-hinted vSIS assumption by [DKLW25a, Theorem 8], and the converse implication follows by [DKLW25a, Corollary 1].*
(5) *Under the stronger strong-hinted vSIS assumption, the same non-blind family also enjoys* sEUF-SMA *by [DKLW25a, Theorem 5].*

Proof. All items are immediate from [DKLW25a, Theorems 4, 5, 7, 8, and Corollary 1] under the stated specialization and side conditions. □

Remark 2.15 (Scope of the DKLW import). Lemma 2.14 is the correct external reduction basis for the non-blind signature layer only. It does not imply blindness or one-more unforgeability of the blind issuance protocol, and it does not by itself yield an ARC theorem.

2.4 Non-interactive zero-knowledge arguments of knowledge (NIZKs)

We use the NIZK-ARK syntax of the SmallWood framework instantiated in the random-oracle model [FR25b, Definition 3].

Definition 2.16 (NIZKs in the random oracle model). Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a family of relations. A (transparent) non-interactive zero-knowledge *argument of knowledge* for $\mathcal{R}$ in the random oracle model (with oracle $\mathsf{RO}(\cdot)$) is a pair of algorithms $(\mathsf{Prove}^{\mathsf{RO}}, \mathsf{Verify}^{\mathsf{RO}})$:

- $\mathsf{Prove}^{\mathsf{RO}}(w, x) \rightarrow \pi$ is probabilistic and on input (x, w) with $(x, w) \in R$ outputs a proof π.
- $\mathsf{Verify}^{\mathsf{RO}}(\pi, x) \rightarrow \{0, 1\}$ is a deterministic polynomial-time algorithm.

Completeness. For all $(x, w) \in \mathcal{R}$,

$$\Pr\left[\mathsf{Verify}^{\mathsf{RO}}(\pi, x) = 1 \;:\; \pi \leftarrow \mathsf{Prove}^{\mathsf{RO}}(w, x)\right] = 1.$$

Straightline-extractable knowledge soundness. There exists a PPT algorithm Ext, also called *extractor*, such that for any PPT adversary $\mathcal{A}$,

$$\Pr\left[\mathsf{Verify}^{\mathsf{RO}}(\pi, x) = 1 \;\wedge\; (x, w) \notin R\right] \;\leq\; \varepsilon,$$

where $(\pi, Q) \leftarrow \mathcal{A}^{\mathsf{RO}}(x)$, Q is the set of query–response pairs made to RO by $\mathcal{A}$, and $w \leftarrow \mathsf{Ext}(Q, \pi)$.

Zero knowledge. There exists a PPT algorithm Sim, called the *simulator*, such that for any $(x, w) \in \mathcal{R}$ and PPT algorithm $\mathcal{A}$:

$$\left|\Pr[\mathcal{A}^{\mathsf{RO}}(\pi_{\mathsf{real}}) = 1] - \Pr[\mathcal{A}^{\mathsf{RO}}(\pi_{\mathsf{sim}}) = 1]\right| \leq \xi,$$

where $\pi_{\mathsf{real}} \leftarrow \mathsf{Prove}^{\mathsf{RO}}(w, x)$, $\pi_{\mathsf{sim}} \leftarrow \mathsf{Sim}^{\mathsf{RO}}(x)$ and Sim can program the output of RO on any input.

We instantiate these proofs using the SmallWood PACS/PIOP framework compiled with Fiat–Shamir in the random oracle model [FR25b]. SmallWood is a commit-and-prove proof system that only relies on symmetric primitives. While its proof size is linear in $|w|$ it is sublinear in the circuit size x.

2.5 Blind signatures

A blind signature algorithm allows a holder $\mathcal{H}$ to obtain signatures valid under the verification key of an issuer $\mathcal{I}$, without the latter learning the signed message μ. Our definitions follow [Fis06]. When any algorithm $\mathcal{B}$ uses the interactive protocol Issue with $\mathcal{I}$, we will write $\mathcal{B}^{\mathsf{Issue}:\langle \mathcal{I}(\cdot):\cdot\rangle}$ and adjust this the notation for other interactions accordingly.

Definition 2.17 (Blind Signature). A Blind Signature scheme is a tuple of algorithms

$$\mathsf{BSig} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Verify})$$

such that:

- $\mathsf{Setup}(1^\lambda)$ is a PPT algorithm that outputs public parameters $\mathsf{pp}_{\mathsf{BSig}}$. All other algorithms obtain $1^\lambda, \mathsf{pp}_{\mathsf{BSig}}$ as implicit inputs.
- $\mathsf{IssuerKeyGen}()$ is a PPT algorithm that outputs issuer keys $(\mathsf{vk}, \mathsf{sk})$.
- Issue is an interactive protocol between $\mathcal{I}$ and $\mathcal{H}$, where $\mathcal{I}$ has input sk and $\mathcal{H}$ has input μ, vk. At the end of the protocol, $\mathcal{H}$ obtains sig.
- $\mathsf{Verify}(\mathsf{vk}, \mu, \mathsf{sig})$ is a deterministic polynomial-time algorithm that outputs a bit.

Definition 2.18 (Correctness). A blind signature scheme is correct if there exists negl such that for all messages μ,

$$\Pr\left[\begin{array}{l} \mathsf{pp}_{\mathsf{BSig}} \leftarrow \mathsf{Setup}(1^\lambda); \\ (\mathsf{vk}, \mathsf{sk}) \leftarrow \mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}}); \\ \mathsf{sig} \leftarrow \mathcal{H}^{\mathsf{Issue}:\langle \mathcal{I}(\mathsf{sk}):\cdot\rangle}(\mu, \mathsf{vk}); \\ \mathsf{Verify}(\mathsf{vk}, \mu, \mathsf{sig}) = 1 \end{array}\right] \geq 1 - \mathsf{negl}(\lambda).$$

Definition 2.19 (Blindness). A blind signature scheme is blind if there exists negl such that for any PPT three-stage stateful adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3)$ and any two messages μ_0, μ_1,

$$\left|\Pr\left[\begin{array}{l} \mathsf{pp}_{\mathsf{BSig}} \leftarrow \mathsf{Setup}(1^\lambda); \\ \mathsf{vk} \leftarrow \mathcal{A}_1(\mathsf{pp}_{\mathsf{BSig}}); \\ b \xleftarrow{\$} \{0, 1\}; \\ \mathcal{A}_2^{\mathsf{Issue}:\langle\cdot:\mathcal{H}(\mu_b, \mathsf{vk})\rangle,\ \mathsf{Issue}:\langle\cdot:\mathcal{H}(\mu_{1-b}, \mathsf{vk})\rangle}; \\ \mathcal{H}(\mu_b, \mathsf{vk}) \text{ outputs } \mathsf{sig}_b; \\ \mathcal{H}(\mu_{1-b}, \mathsf{vk}) \text{ outputs } \mathsf{sig}_{1-b} \\ \text{if } \perp \in \{\mathsf{sig}_0, \mathsf{sig}_1\} \text{ then } (\mathsf{sig}_0, \mathsf{sig}_1) \leftarrow (\perp, \perp) : \\ b = \mathcal{A}_3(\mathsf{sig}_0, \mathsf{sig}_1) \end{array}\right] - \frac{1}{2}\right| \leq \mathsf{negl}(\lambda).$$

Definition 2.20 (One-more unforgeability). A blind signature scheme is one-more unforgeable if there exists negl such that for any PPT adversary $\mathcal{A}^{\mathsf{Issue}:\langle \mathcal{I}(\mathsf{sk}):\cdot\rangle}(\mathsf{vk})$ that makes at most $Q = Q(\lambda)$ oracle queries to Issue,

$$\Pr\left[\begin{array}{l} \mathsf{pp}_{\mathsf{BSig}} \leftarrow \mathsf{Setup}(1^\lambda); \\ (\mathsf{vk}, \mathsf{sk}) \leftarrow \mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}}); \\ \{(\mu_i, \mathsf{sig}_i)\}_{i \in [Q+1]} \leftarrow \mathcal{A}^{\mathsf{Issue}\langle \mathcal{I}(\mathsf{sk}):\cdot\rangle}(\mathsf{vk}) : \\ \forall\, 1 \leq i < j \leq Q + 1 :\; \mu_i \neq \mu_j \;\wedge\; \forall\, i \in [Q + 1] :\; \mathsf{Verify}(\mathsf{vk}, \mu_i, \mathsf{sig}_i) = 1 \end{array}\right] \leq$$

Rate limiting at the ARC layer. We do not introduce a separate stand-alone rate-limiting primitive. In the present paper, rate limiting is part of the ARC syntax itself: the holder presents a credential together with a public nonce and a PRF-derived public tag, and the verifier enforces the policy through local state on previously accepted (nonce, tag) pairs.

2.6 Anonymous Rate-Limiting Credentials (ARCs)

Anonymous rate-limited credentials extend blind signatures by adding a showing algorithm Show that may create multiple presentations of a previously issued credential. We explicitly introduce the verifier party $\mathcal{V}$, which receives and checks those presentations. Our definition follows the established syntax and security models of [FHS19, BLNS23b], specialized to the ARC setting considered here.

We keep Setup and IssuerKeyGen separate, so the public commitment matrix is sampled independently of the issuer keys. We also separate Show and Verify to distinguish the client-side presentation algorithm from the verifier-side stateful check.

Definition 2.21 (ARC scheme syntax)**.** Let $\mathcal{N}$ be a finite set of nonces. An anonymous rate-limited credential (ARC) scheme is a tuple of algorithms

$$\mathsf{ARC} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Show}, \mathsf{Verify})$$

such that:

- $\mathsf{Setup}(1^\lambda)$ is a PPT algorithm that outputs public parameters $\mathsf{pp}_{\mathsf{ARC}}$. All other algorithms obtain $1^\lambda, \mathsf{pp}_{\mathsf{ARC}}$ as implicit inputs.
- $\mathsf{IssuerKeyGen}()$ is a PPT algorithm that outputs issuer keys $(\mathsf{vk}, \mathsf{sk})$.
- $\mathsf{Issue}(\cdot)$ is an interactive protocol between $\mathcal{I}$ and $\mathcal{H}$, where $\mathcal{I}$ has input sk and $\mathcal{H}$ has input vk. At the end of the protocol, $\mathcal{H}$ obtains cred.
- $\mathsf{Show}(\cdot)$ on input a credential cred and a nonce nonce $\in \mathcal{N}$ outputs a presentation pres.
- $\mathsf{Verify}(\mathsf{vk}, \mathsf{pres}, \mathsf{nonce})$ is a deterministic polynomial-time algorithm that outputs a bit.

Adjusting the correctness definition 2.18, blindness definition 2.19 and unforgeability definition 2.20 to ARC follows trivially. However, we additionally require that if the same credential cred is presented multiple times, but for different nonces from $\mathcal{N}$, then the presentations should be unlinkable. To formalize this, we let a challenger C create two credentials in interactions where the adversary controls the issuer. Then the challenger either creates two presentations for the same credential, or for different credentials.

Definition 2.22 (Presentation unlinkability)**.** Let ARC be an ARC scheme. Consider the following experiment $\mathsf{Exp}^{\text{pres-unlink}}_{\mathsf{ARC},\mathcal{A}}(\lambda)$ between a challenger C and a two-stage stateful adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$:

(1) C runs $\mathsf{pp}_{\mathsf{ARC}} \leftarrow \mathsf{Setup}(1^\lambda)$, $(\mathsf{vk}, \mathsf{sk}) \leftarrow \mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{ARC}})$.
(2) Run $\mathcal{A}_1^{\mathsf{Issue}:\langle\cdot:C(\mathsf{vk})\rangle,\ \mathsf{Issue}:\langle\cdot:C(\mathsf{vk})\rangle}$, at the end of which C obtains $\mathsf{cred}_0, \mathsf{cred}_1$ and $\mathcal{A}_1$ outputs $\mathsf{nonce}_0, \mathsf{nonce}_1$.
(3) If cred_0 or cred_1 is $\perp$ or if $\mathsf{nonce}_0 = \mathsf{nonce}_1$ then output a random bit.
(4) C samples $b \xleftarrow{\$} \{0,1\}$ and computes

$$\mathsf{pres}_0 \leftarrow \mathsf{Show}(\mathsf{cred}_0, \mathsf{nonce}_0), \qquad \mathsf{pres}_1 \leftarrow \mathsf{Show}(\mathsf{cred}_b, \mathsf{nonce}_1).$$

(5) $\mathcal{A}_2(\mathsf{pres}_0, \mathsf{pres}_1)$ outputs a bit b'.

Define $\mathcal{A}^{\text{pres-unlink}}_{\mathsf{ARC}}(\mathcal{A}) := \left|\Pr[b' = b] - \frac{1}{2}\right|$. We say ARC has *presentation unlinkability* if for all PPT $\mathcal{A}$ this advantage is negligible in λ.

2.7 Pseudorandom functions

Definition 2.23 (Pseudorandom function (PRF))**.** Let $\mathcal{K}$, $\mathcal{D}$ and $\mathcal{T}$ be sets such that $|\mathcal{K}|$ is exponential in λ. A function family $\mathsf{F} : \mathcal{K} \times \mathcal{D} \to \mathcal{T}$ is a PRF if for all PPT adversaries $\mathcal{A}$,

$$\mathcal{A}^{\mathrm{prf}}_{\mathsf{F}}(\mathcal{A}) := \left|\Pr[\mathcal{A}^{O_0} = 1] - \Pr[\mathcal{A}^{O_1} = 1]\right| \le \mathrm{negl}(\lambda),$$

where $O_0(\cdot) = \mathsf{F}(k, \cdot)$ for uniform $k \xleftarrow{\$} \mathcal{K}$ and $O_1(\cdot)$ is a uniform random function from $\mathcal{D}$ to $\mathcal{T}$.

In this work, we instantiate PRFs based on the Poseidon2 construction [GKS23].

It is based on the Poseidon2 permutation, which is defined as follows:

Definition 2.24 (Poseidon2-style permutation structure)**.** Let q be a prime, $d \ge 3$ such that $\gcd(d, q-1) = 1$ and $t, R_F, R_P \in \mathbb{N}$ where R_F is even. We call R_F the *external* rounds and R_P the *internal* rounds. Furthermore, let $\{c_j^{(i)}\}_{i\in[R_F]}^{j\in[t]} \subset \mathbb{F}_q$, $\{\hat{c}^{(i)}\}_{i\in[R_P]} \subset \mathbb{F}_q$ be public round constants and $\mathbf{M}_E, \mathbf{M}_I \in \mathbb{F}_q^{t\times t}$ be invertible maps.

For $\mathbf{x} = (x_1, \ldots, x_t) \in \mathbb{F}_q^t$, define external rounds

$$E_i(\mathbf{x}) := \mathbf{M}_E \cdot \left((x_1 + c_1^{(i)})^d, \ldots, (x_t + c_t^{(i)})^d\right)^\top,$$

and internal rounds

$$I_i(\mathbf{x}) := \mathbf{M}_I \cdot \left((x_1 + \hat{c}^{(i)})^d,\ x_2, \ldots, x_t\right)^\top.$$

Then

$$P := E_{R_F} \circ \cdots \circ E_{R_F/2+1} \circ I_{R_P} \circ \cdots \circ I_1 \circ E_{R_F/2} \circ \cdots \circ E_1.$$

We additionally define $\mathrm{Tr}_i : \mathbb{F}_q^t \to \mathbb{F}_q^i$ outputs the first i elements of the input vector.

Definition 2.25 (Poseidon2 PRF)**.** Let $\ell_{\mathsf{tag}}, \ell_{\mathsf{key}}, \ell_{\mathsf{in}} \in \mathbb{N}$ and set $t = \ell_{\mathsf{key}} + \ell_{\mathsf{nonce}}$. For a key $\mathbf{k} \in \mathbb{F}_q^{\ell_{\mathsf{key}}}$ and public input $\mathbf{n} \in \mathbb{F}_q^{\ell_{\mathsf{in}}}$, we define the Poseidon2 PRF as

$$\mathsf{F}(\mathbf{k}, \mathbf{n}) := \mathrm{Tr}_{\ell_{\mathsf{tag}}}\left(P(\mathbf{k}\|\mathbf{n}) + (\mathbf{k}\|\mathbf{n})\right),$$

where $P : \mathbb{F}_q^t \to \mathbb{F}_q^t$ is the permutation from Definition 2.24.

Constraint view (what the post-sign NIZK must prove). To prove $\mathsf{tag} = \mathsf{F}(k, \mathsf{nonce})$ inside SmallWood, where k is the signed secret part of $m = \mu\|k$, we arithmetize: (i) S-box constraints of the form $z = y^d$ at selected checkpoints, (ii) linear constraints for adding round constants and applying M_E, M_I between checkpoints, and (iii) feed-forward plus truncation constraints binding the public tag to the final state. The constraint families are stated in Section 5.5, while checkpoint scheduling and extended notes are in Appendix E.

3 BLIND SIGNATURE CONSTRUCTION

This section presents the vSIS-based blind signature that underlies ARC-SPRUCE. It combines the Ajtai commitment of Section 2.2, the trapdoor rational hash of Definition 2.7, and the NIZK-ARK of Definition 2.16. The signed message is always

$$\mathbf{m} = \mu\|\mathbf{k},$$

where μ denotes the attribute payload and $\mathbf{k}$ is a holder-generated secret PRF key used later for rate limiting.

3.0.1 Randomizing the rational-hash witness. In the protocol, $\mathcal{H}$ first commits to

$$\mathbf{w} = [\mathbf{m}\|\mathbf{r}_0^{\mathcal{H}}\|\mathbf{r}_1^{\mathcal{H}}\|\bar{\mathbf{r}}]^\top,$$

where $(\mathbf{r}_0^{\mathcal{H}}, \mathbf{r}_1^{\mathcal{H}})$ are the holder contributions to the rational-hash witness and $\bar{\mathbf{r}}$ is fresh commitment randomness. To satisfy the commitment bounds, we require $\|\mathbf{w}\|_\infty \le \beta_{\mathsf{msg}}$.

The issuance process then uses a two-party randomness generation step: the issuer samples $(\mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}})$, and the joint witness is obtained coefficient-wise by

$$\mathbf{r}_0 = \mathsf{center}(\mathbf{r}_0^{\mathcal{H}} + \mathbf{r}_0^{\mathcal{I}}), \qquad \mathbf{r}_1 = \mathsf{center}(\mathbf{r}_1^{\mathcal{H}} + \mathbf{r}_1^{\mathcal{I}}).$$

Appendix A records the centering map and its uniformity property.

3.0.2 Blind signature protocol. We assume $\mathsf{NIZK} = (\mathsf{Prove}, \mathsf{Verify})$ is a NIZK for the relations below, $\mathsf{Com} = (\mathsf{Setup}, \mathsf{Commit}, \mathsf{Verify})$ is the Ajtai commitment scheme, and

$$\mathsf{TDRH} = (\mathsf{Setup}, \mathsf{TrapGen}, \mathsf{Eval}, \mathsf{SampPre})$$

is the trapdoor rational-hash interface from Definition 2.6. Public setup outputs the hash key B, while issuer key generation outputs the trapdoored signing matrix $\mathbf{A}$.

Abstract and compiled issuance relations. We distinguish the ideal issuance relation from the relation actually passed to the pre-sign proof system.

We write $R^{\mathrm{abs}}_{\mathrm{Issue}}(x, w) = 1$ iff

$$\begin{aligned}
x &= \big(\mathsf{pp}_{\mathsf{Com}}, \mathbf{c}, T, (\mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}}), B\big),\\
w &= (\mathbf{m}, \mathbf{r}_0^{\mathcal{H}}, \mathbf{r}_1^{\mathcal{H}}, \bar{\mathbf{r}}),\\
\mathsf{pp}_{\mathsf{Com}} &= (\mathbf{A}_{\mathsf{Com}}, \beta_{\mathsf{msg}}),\\
\mathbf{c} &= \mathbf{A}_{\mathsf{Com}} \cdot [\mathbf{m}\|\mathbf{r}_0^{\mathcal{H}}\|\mathbf{r}_1^{\mathcal{H}}\|\bar{\mathbf{r}}]^\top,\\
\|\mathbf{m}\|_\infty, \|\mathbf{r}_0^{\mathcal{H}}\|_\infty, \|\mathbf{r}_1^{\mathcal{H}}\|_\infty, \|\bar{\mathbf{r}}\|_\infty &\le \beta_{\mathsf{msg}},\\
\mathbf{r}_0 &= \mathsf{center}(\mathbf{r}_0^{\mathcal{H}} + \mathbf{r}_0^{\mathcal{I}}),\\
\mathbf{r}_1 &= \mathsf{center}(\mathbf{r}_1^{\mathcal{H}} + \mathbf{r}_1^{\mathcal{I}}),\\
T &= h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B).
\end{aligned}$$

We write $R^{\mathrm{cmp}}_{\mathrm{Issue}}(x, w) = 1$ iff the same conditions hold, except that the last line is replaced by

$$(B_3 - \mathbf{r}_1) \odot T - \big(B_0 + B_1 \cdot \mathbf{m} + B_2 \cdot \mathbf{r}_0\big) = 0.$$

The issued signature relation remains the abstract relation

$$R_{\mathrm{Sign}}(x, w) = 1 \iff x = (\mathbf{m}, \mathbf{A}, B, \beta_{\mathsf{msg}}, \beta_{\mathsf{sig}}), \quad w = (\mathbf{u}, \mathbf{r}_0, \mathbf{r}_1),$$

$$\|\mathbf{r}_0\|_\infty, \|\mathbf{r}_1\|_\infty \le \beta_{\mathsf{msg}}, \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathsf{sig}},$$

$$h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B) \ne \bot, \qquad \mathbf{A}\mathbf{u} = h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B).$$

Lemma 3.1. *If an honest holder first checks that $B_3 - \mathbf{r}_1$ is invertible in every CRT slot and then computes $T = h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B)$, every witness satisfying $R^{\mathrm{abs}}_{\mathrm{Issue}}$ also satisfies $R^{\mathrm{cmp}}_{\mathrm{Issue}}$. The converse need not hold.*

Proof. This is immediate from Lemma 2.8 and Remark 2.9. □

We now define the blind-signature algorithms.

$\mathsf{Setup}(1^\lambda)$**:** Fix the global ring parameters. Run

$$(\mathbf{A}_{\mathsf{Com}}, \beta_{\mathsf{msg}}) \leftarrow \mathsf{Com.Setup}(1^\lambda)$$

and

$$(B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\mathsf{msg}}, \beta_{\mathsf{sig}}) \leftarrow \mathsf{TDRH.Setup}(1^\lambda).$$

Output

$$\mathsf{pp}_{\mathsf{BSig}} = (\mathbf{A}_{\mathsf{Com}}, B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\mathsf{msg}}, \beta_{\mathsf{sig}}).$$

$\mathsf{IssuerKeyGen}()$**:** Compute $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TDRH.TrapGen}(1^\lambda)$ and output

$$\mathsf{vk} = \mathbf{A}, \qquad \mathsf{sk} = \mathsf{td}.$$

Issue**:** The issuer $\mathcal{I}$ holds sk; the holder $\mathcal{H}$ holds μ, vk.

(1) $\mathcal{H}$ samples $\mathbf{k}, \mathbf{r}_0^{\mathcal{H}}, \mathbf{r}_1^{\mathcal{H}}, \bar{\mathbf{r}}$ coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$ and sets $\mathbf{m} = \mu\|\mathbf{k}$.

(2) $\mathcal{H}$ computes

$$\mathbf{c} \leftarrow \mathbf{A}_{\mathsf{Com}} \cdot [\mathbf{m}\|\mathbf{r}_0^{\mathcal{H}}\|\mathbf{r}_1^{\mathcal{H}}\|\bar{\mathbf{r}}]^\top$$

and sends $\mathbf{c}$ to $\mathcal{I}$.

(3) $\mathcal{I}$ samples $\mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}}$ coefficient-wise in $[-\beta_{\mathsf{msg}}, \beta_{\mathsf{msg}}]$ and sends them to $\mathcal{H}$.

(4) $\mathcal{H}$ computes

$$\mathbf{r}_0 \leftarrow \mathsf{center}(\mathbf{r}_0^{\mathcal{H}} + \mathbf{r}_0^{\mathcal{I}}), \qquad \mathbf{r}_1 \leftarrow \mathsf{center}(\mathbf{r}_1^{\mathcal{H}} + \mathbf{r}_1^{\mathcal{I}}),$$

checks whether $h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B) = \bot$, and aborts/restarts if so. Otherwise it computes the public target

$$T \leftarrow h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B).$$

(5) $\mathcal{H}$ computes $\pi_{\mathsf{Issue}} \leftarrow \mathsf{NIZK.Prove}(w, x)$ for the compiled relation $R^{\mathrm{cmp}}_{\mathrm{Issue}}$, with witness

$$w = (\mathbf{m}, \mathbf{r}_0^{\mathcal{H}}, \mathbf{r}_1^{\mathcal{H}}, \bar{\mathbf{r}})$$

and statement

$$x = (\mathsf{pp}_{\mathsf{Com}}, \mathbf{c}, T, (\mathbf{r}_0^{\mathcal{I}}, \mathbf{r}_1^{\mathcal{I}}), B),$$

and sends $(T, \pi_{\mathsf{Issue}})$ to $\mathcal{I}$.

(6) $\mathcal{I}$ verifies π_{Issue} for $R^{\mathrm{cmp}}_{\mathrm{Issue}}$. If verification fails, it aborts. Otherwise it computes

$$\mathbf{u} \leftarrow \mathsf{TDRH.SampPre}(\mathsf{td}, T)$$

and sends $\mathbf{u}$ to $\mathcal{H}$.

(7) $\mathcal{H}$ checks $\|\mathbf{u}\|_\infty \le \beta_{\mathsf{sig}}$ and $\mathbf{A}\mathbf{u} = T$. If so, it outputs

$$\mathsf{sig} = (\mathbf{u}, \mathbf{r}_0, \mathbf{r}_1)$$

on message $\mathbf{m} = \mu\|\mathbf{k}$; otherwise it aborts.

$\mathsf{Verify}(\mathsf{vk}, \mathbf{m}, \mathsf{sig})$**:** Let $\mathsf{vk} = \mathbf{A}$. Parse $\mathsf{sig} = (\mathbf{u}, \mathbf{r}_0, \mathbf{r}_1)$. Accept iff

$$\big((\mathbf{m}, \mathbf{A}, B, \beta_{\mathsf{msg}}, \beta_{\mathsf{sig}}), (\mathbf{u}, \mathbf{r}_0, \mathbf{r}_1)\big) \in R_{\mathrm{Sign}}.$$

For ARC, the holder does not reveal sig directly. Section 4 instead proves the compiled post-sign relation of Section 5 in zero knowledge while keeping $\mathbf{k}, \mathbf{r}_0, \mathbf{r}_1$ hidden and binding $\mathbf{k}$ to the PRF tag.

3.0.3 Security.

Proposition 3.2 (Correctness)**.** *Assume the pre-sign NIZK is complete and the trapdoor sampler is correct. Then an honest execution of* Issue *outputs a signature* sig *that is accepted by* Verify*, except with negligible failure probability coming from the trapdoor sampler or an inadmissible hash evaluation.*

Proof. By honest target derivation, the holder computes T only after checking that $h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B) \neq \perp$, and then sets

$$T = h_{\mathbf{m},(\mathbf{r}_0,\mathbf{r}_1)}(B).$$

Lemma 3.1 implies that the honest witness satisfies the compiled relation $R^{\text{cmp}}_{\text{Issue}}$, so completeness of the pre-sign NIZK implies that the issuer accepts. By correctness of SampPre, the returned $\mathbf{u}$ satisfies $\mathbf{Au} = T$ and the public shortness bound except with negligible probability. Hence sig $= (\mathbf{u}, \mathbf{r}_0, \mathbf{r}_1)$ satisfies R_{Sign}. □

Remark 3.3 (Security status of the blind protocol). The non-blind hash-and-preimage signature layer underlying this protocol is justified by the DKLW25 reduction summarized in Lemma 2.14. For the blind protocol itself, the only theorem-backed property proved in the present draft is honest-party correctness. A proof of one-more unforgeability would still have to combine the extractability of the pre-sign proof, the hiding of the Ajtai commitment, the algebraic correctness of the centering step, and an interactive one-more hardness statement that genuinely matches the issuance oracle. A proof of blindness would still have to combine the same commitment-hiding and proof zero-knowledge ingredients with a simulation argument for the issuer view. Accordingly, the blind protocol is presented here as a construction with defined goals and an explicit reduction roadmap, not as a layer for which blindness or one-more unforgeability has already been reduced.

4 ARC CONSTRUCTION FROM BLIND SIGNATURES

This section turns the blind signature of Section 3 into an anonymous rate-limited credential (ARC) system. We use the following terminology throughout: *issuance* refers to the blind signature protocol, while *showing* refers to an ARC presentation. The issuance proof is the *pre-sign NIZK*; the showing proof is the *post-sign NIZK*.

4.1 From blind signatures to ARC

A credential issued to a holder conceptually consists of:

- the signed message $m = \mu \| \mathbf{k}$, where μ are attributes and $\mathbf{k}$ is a uniformly sampled secret PRF key;
- the signature u satisfying $Au = h_{m,(r_0,r_1)}(B)$ for some bounded randomness (r_0, r_1).

The holder may additionally store local auxiliary values needed to prove the signature relation in zero knowledge. These values are never revealed in showings.

Message structure. Every credential message has the abstract form

$$m = \mu \| \mathbf{k}.$$

Here μ encodes the holder attributes and $\mathbf{k}$ is sampled uniformly and serves as the secret PRF key used for rate limiting. At the proof-system layer this abstract message is represented by packed witness rows (M, K), but the credential syntax itself remains the quotient-ring statement of Section 3.

4.2 Rate limiting via the secret key k

Rate limiting is implemented by deterministic tags derived from a PRF keyed by the credential's secret $\mathbf{k}$. For a public nonce nonce $\in \mathcal{D}$, the holder computes

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

Intuitively:

- if the holder never repeats a nonce, then tag is pseudorandom and yields unlinkable showings;
- if the holder repeats a nonce, the tag repeats deterministically, enabling the verifier to detect double-spending for that nonce.

The verifier enforces its rate policy by restricting acceptable nonces and maintaining state that records previously accepted (nonce, tag) pairs for a given policy context.

4.3 Concrete PRF instantiation (Poseidon2-like)

We instantiate the PRF from Definition 2.23 using the Poseidon2-like construction of Definitions 2.25 and 2.24. Algorithm 1 summarizes the computation.

Algorithm 1 $\mathsf{F}(k, n)$ (Poseidon2-like with feed-forward + truncation)

$x \leftarrow k \| n$; $x^{(0)} \leftarrow x$
for $i = 1$ to $R_F/2$ **do**
  $x \leftarrow E_i(x)$
for $i = 1$ to R_P **do**
  $x \leftarrow I_i(x)$
for $i = R_F/2 + 1$ to R_F **do**
  $x \leftarrow E_i(x)$
$y \leftarrow x + x^{(0)}$ ▷ feed-forward
return $\mathrm{Tr}(y)$ ▷ first ℓ_{tag} lanes

Proof-friendly arithmetization. In the post-sign NIZK we enforce $\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$ using a checkpointed arithmetization: we commit selected S-box outputs and let the verifier replay all linear wiring between checkpoints from openings and public constants. Section 5.5 states the constraint families; Appendix E records the checkpoint schedule and extended notes.

4.4 Showing protocol

Algorithm 2 Show: ARC presentation (canonical protocol)

Require: Holder has credential $(\mu, \mathbf{k}, r_0, r_1, u)$. Public parameters include $A, B, \beta_{\mathsf{msg}}$ and the PRF description.
Require: Input nonce nonce $\in \mathcal{D}$ that has not been used before for this credential.
Ensure: Presentation $(\mathsf{tag}, \mathsf{nonce}, \pi_{\mathsf{post}})$.
1: Compute tag $\leftarrow \mathsf{F}(\mathbf{k}, \mathsf{nonce})$.
2: Produce a post-sign NIZK π_{post} for the compiled showing relation specified in Section 5, with public inputs $(\mathsf{tag}, \mathsf{nonce})$ and witness derived from the credential $(\mu, \mathbf{k}, r_0, r_1, u)$.
3: Output $(\mathsf{tag}, \mathsf{nonce}, \pi_{\mathsf{post}})$.

At the ARC level, the intended abstract witness remains $(u, \mu, \mathbf{k}, r_0, r_1)$ satisfying the signature equation and the PRF-tag equation. Section 5 specifies the compiled SmallWood relation used to realize that goal at the proof-system layer and proves its coefficient-domain consequence.

The verifier runs Verify by checking π_{post} and consulting a policy state that tracks used nonces and tags. The verifier rejects if the proof fails or if the nonce-tag pair violates the local policy.

4.5 PRF role and rationale

The PRF serves two simultaneous roles:

- **Public rate-limiting handle.** The tag is deterministic in $(\mathbf{k}, \mathsf{nonce})$ so the verifier can enforce stateful policies without interacting with the issuer and without learning the credential message.
- **Unlinkability across nonces.** For fresh nonces, tags are pseudorandom even given previous tags, so a verifier cannot link two presentations to the same credential unless the policy context forces reuse.

Crucially, $\mathbf{k}$ is part of the signed message, so the holder cannot change the PRF key between showings without producing a new valid signature witness. The post-sign NIZK binds the public tag to that signed hidden key while keeping the key itself hidden. This is the mechanism by which verifier-side state can enforce rate policies through the public pair $(\mathsf{nonce}, \mathsf{tag})$.

4.6 ARC correctness and conditional security decomposition

Honest correctness. If issuance completes, the holder stores a witness $(u, \mu, \mathbf{k}, r_0, r_1)$ satisfying the abstract signature relation of Section 3. For any public nonce nonce, the holder computes tag $= \mathsf{F}(\mathbf{k}, \mathsf{nonce})$ and derives the corresponding witness for the compiled post-sign relation of Section 5. By completeness of the post-sign NIZK, the verifier accepts any honest presentation for which the local policy check on $(\mathsf{nonce}, \mathsf{tag})$ also accepts.

Cryptographic decomposition of the ARC layer. The ARC layer combines five ingredients:

(1) the non-blind theorem-backed signature layer of Lemma 2.14;
(2) blindness and one-more unforgeability of the blind issuance protocol, which remain to be reduced;
(3) straightline-extractable knowledge soundness and zero knowledge of the post-sign NIZK for the compiled relation of Section 5;
(4) pseudorandomness and correctness of the PRF; and
(5) verifier-side policy state recording previously accepted $(\mathsf{nonce}, \mathsf{tag})$ pairs.

Only the first and third ingredients are theorem-backed in the present draft. The fourth is an external assumption on the chosen PRF instantiation, the second remains a reduction goal, and the fifth is an application-layer enforcement rule rather than a stand-alone cryptographic theorem.

Conditional policy soundness. Knowledge soundness of the post-sign proof yields a witness for the compiled showing relation of Section 5. By Corollary 5.2, every accepting post-sign proof yields

$$B_3Au - (Au)R_1 - B_0 - B_1(M + K) - B_2R_0 = 0$$

in R_q. If the extracted witness additionally satisfies denominator admissibility, then Corollary 5.1 yields the abstract signature relation

$$Au = h_{M+K,(R_0,R_1)}(B).$$

Under that signature-side condition, together with one-more unforgeability of the blind issuance layer and correctness of the PRF, the signed key k determines the public tag uniquely for each nonce. Consequently, a verifier that records spent $(\mathsf{nonce}, \mathsf{tag})$ pairs rejects same-nonce replays deterministically.

Conditional presentation unlinkability. Presentation unlinkability in Definition 2.22 is not implied by PRF security and post-sign zero knowledge alone. In that experiment the adversary controls the issuer during the two issuance sessions. Accordingly, unlinkability also depends on an issuance-layer privacy property, namely blindness or a comparably strong issuer-view hiding statement. Under that issuance-layer privacy assumption, post-sign zero knowledge hides the witness used during showing, and PRF pseudorandomness makes the public tags computationally random across distinct fresh nonces. This yields the intended unlinkability claim for showing transcripts, subject to the explicit nonce-reuse caveat.

Attribute privacy and selective disclosure. The message part μ can encode attributes; the post-sign NIZK can be extended to prove predicates about μ and to selectively reveal components, without changing the issuance protocol. We focus on the core rate-limiting construction and defer predicate engineering to the SmallWood programming model in Section 5.

5 SMALLWOOD PROGRAMMING MODEL

This section specifies the SmallWood relations used by ARC-SPRUCE. Its purpose is to record the committed witness surfaces, public inputs, and PACS constraint families used by the issuance and showing proofs, and to state the precise strength of the resulting proof statements. Throughout, SmallWood certifies only the compiled relations stated in this section, namely relations over committed witness rows, public lifts, and PACS constraint families derived from them. Any further coefficient-domain consequence is obtained only through the explicit theorem stated below.

5.1 Proof-language interface

Fix an evaluation set $\Omega = \{\omega_1, \ldots, \omega_s\} \subset \mathbb{F}_q$ and a disjoint set of blinding points $\Omega' \subset \mathbb{F}_q \setminus \Omega$. A witness is arranged as a matrix

$$W \in \mathbb{F}_q^{n \times s},$$

whose rows are interpolated into low-degree polynomials

$$P_1(X), \ldots, P_n(X) \in \mathbb{F}_q[X]$$

satisfying

$$P_i(\omega_k) = W_{i,k} \qquad \text{for all } \omega_k \in \Omega,$$

and taking random values on Ω' in order to achieve honest-verifier zero knowledge. SmallWood commits to these row polynomials and proves satisfaction of PACS constraints by opening verifier-chosen linear maps of the committed rows.

The PACS language distinguishes two classes of constraints.

- *Parallel constraints* are enforced pointwise on Ω. They are used for slot-local algebra, including linear relations, pointwise products, selector gating, and checkpointed PRF constraints.
- *Aggregated constraints* are enforced through Ω-sum identities. They are used when the encoding introduces global coupling across packed coordinates, in particular for public linear bridge relations between source rows and replay-image rows.

The proof-system properties imported from SmallWood apply to the compiled PACS relation built from these rows and constraints. They do not, by themselves, replace the explicit coefficient-domain consequence theorem stated in Section 5.6.

5.2 Abstract and compiled relations

At the protocol layer, the blind-signature and ARC constructions are specified by abstract relations in the quotient ring R_q. The SmallWood proofs, however, are proofs for compiled relations over committed witness rows, public lifts, and PACS constraints. We therefore keep the two levels distinct throughout this section.

In issuance, the compiled relation is the cleared-denominator relation $R^{\mathrm{cmp}}_{\mathrm{Issue}}$ of Section 3, with the public target T supplied as an input to the proof. In showing, the compiled relation is a full transform-basis replay relation over explicitly committed source rows, full replay-image blocks, and PRF companion rows. The purpose of the remainder of the section is to state those compiled relations precisely and to prove their coefficient-domain consequence.

5.3 Issuance (pre-sign) compiled relation

The pre-sign proof keeps the blind-signature target T public and proves that it is consistent with one hidden commitment opening and the issuer randomness. At the abstract level, issuance is governed by $R^{\mathrm{abs}}_{\mathrm{Issue}}$ from Section 3. The compiled SmallWood proof certifies instead the weaker relation $R^{\mathrm{cmp}}_{\mathrm{Issue}}$, which replaces the defined quotient semantics by the cleared-denominator relation.

Committed rows. The committed pre-sign witness consists of the following rows:

$$C^M,\ C^{\mathrm{preRU}},\ C^{\mathrm{preR}},\ C^{\mathrm{ctr}},\ C^J,$$
$$M,\ K,\ R_0^U,\ R_1^U,\ R,\ R_0,\ R_1,\ J_0,\ J_1,$$
$$\widehat{M},\ \widehat{K},\ \widehat{R}_0,\ \widehat{R}_1.$$

The first five rows are low-alphabet carriers. The next nine rows are decoded aliases. The final four rows are the transform aliases used by the hash replay.

Public inputs and lifts. The public pre-sign inputs are

$$\mathbf{A}_{\mathrm{Com}},\ B,\ \mathbf{c},\ \mathbf{r}_0^{\mathcal{I}},\ \mathbf{r}_1^{\mathcal{I}},\ T.$$

The verifier also derives the public lifts

$$B_0^\Theta,\ B_1^\Theta,\ B_2^\Theta,\ B_3^\Theta,\ T^\Theta,$$

where T^Θ is the chosen public lift of the coefficient-domain target T.

Constraint families. Let $\Delta = 2\beta_{\mathsf{msg}} + 1$. The compiled pre-sign relation enforces:

(1) *Carrier membership:*

$$P_M(C^M) = 0, \qquad P_{\mathrm{preRU}}(C^{\mathrm{preRU}}) = 0, \qquad P_{\mathrm{preR}}(C^{\mathrm{preR}}) = 0,$$
$$P_{\mathrm{ctr}}(C^{\mathrm{ctr}}) = 0, \qquad P_J(C^J) = 0.$$

(2) *Carrier-to-alias bridges:*

$$M - \mathrm{Dec}_M^{(1)}(C^M) = 0, \qquad K - \mathrm{Dec}_M^{(2)}(C^M) = 0,$$
$$R_0^U - \mathrm{Dec}_{\mathrm{preRU}}^{(1)}(C^{\mathrm{preRU}}) = 0, \qquad R_1^U - \mathrm{Dec}_{\mathrm{preRU}}^{(2)}(C^{\mathrm{preRU}}) = 0,$$
$$R - \mathrm{Dec}_{\mathrm{preR}}^{(1)}(C^{\mathrm{preR}}) = 0,$$
$$R_0 - \mathrm{Dec}_{\mathrm{ctr}}^{(1)}(C^{\mathrm{ctr}}) = 0, \qquad R_1 - \mathrm{Dec}_{\mathrm{ctr}}^{(2)}(C^{\mathrm{ctr}}) = 0,$$
$$J_0 - \mathrm{Dec}_J^{(1)}(C^J) = 0, \qquad J_1 - \mathrm{Dec}_J^{(2)}(C^J) = 0.$$

(3) *Commitment relation:*

$$\mathbf{A}_{\mathrm{Com}} \cdot [M \| K \| R_0^U \| R_1^U \| R]^\top = \mathbf{c}.$$

(4) *Centering relation:*

$$R_0 = R_0^U + \mathbf{r}_0^{\mathcal{I}} - \Delta J_0, \qquad R_1 = R_1^U + \mathbf{r}_1^{\mathcal{I}} - \Delta J_1.$$

(5) *Packing relation:* the rows M and K occupy complementary halves of the packing domain and together encode the abstract message $m = \mu \| k$.

(6) *Transform-domain cleared relation:*

$$(B_3^\Theta - \widehat{R}_1) \odot T^\Theta - \big(B_0^\Theta + B_1^\Theta \odot (\widehat{M} + \widehat{K}) + B_2^\Theta \odot \widehat{R}_0\big) = 0.$$

Aggregated bridge families. The issuance statement includes the four transform bridges

$$M \mapsto \widehat{M}, \qquad K \mapsto \widehat{K}, \qquad R_0 \mapsto \widehat{R}_0, \qquad R_1 \mapsto \widehat{R}_1.$$

Statement proved. The compiled pre-sign proof asserts the existence of committed carriers, decoded aliases, and transform aliases such that:

(1) each carrier lies in its designated carrier alphabet;
(2) each decoded alias is the decode image of its carrier;
(3) the decoded aliases satisfy commitment, centering, and packing;
(4) the transform aliases are the public transform images of the decoded rows; and
(5) the public target T, after lifting, satisfies the transform-domain cleared relation.

This is a proof of $R^{\mathrm{cmp}}_{\mathrm{Issue}}$. No inverse witness for $B_3 - R_1$ is included, and the compiled statement should therefore not be described as a proof of the abstract quotient semantics.

5.4 Showing (post-sign) compiled relation

At the ARC layer, the abstract showing relation is the existence of a witness (u, μ, k, r_0, r_1) such that

$$Au = h_{\mu\|k,(r_0,r_1)}(B) \qquad \text{and} \qquad \mathsf{tag} = \mathsf{F}(k, \mathsf{nonce}),$$

together with the public coefficient-wise bounds on μ, k, r_0, r_1 and the public shortness bound on u. The post-sign NIZK does not expose this witness directly. Instead, it proves a compiled transform-basis relation over committed source rows, full replay-image blocks, and PRF companion rows.

Exact replay transform. Let

$$\mathcal{F}_{\mathrm{rep}} : R_q \longrightarrow \mathcal{T}$$

denote the exact public transform used by the replay relation, where $\mathcal{T}$ is the full transform image of R_q for the chosen replay basis. We write

$$\mathcal{F}_{\mathrm{rep}} = \left(\mathcal{F}_{\mathrm{rep},0}, \ldots, \mathcal{F}_{\mathrm{rep},B_{\mathrm{rep}}-1}\right)$$

for its decomposition into replay blocks. For every public hash polynomial $B_i \in R_q$, define the corresponding block lifts

$$B^{\Theta}_{i,b} := \mathcal{F}_{\mathrm{rep},b}(B_i), \qquad i \in \{0, 1, 2, 3\}, \quad b \in \{0, \ldots, B_{\mathrm{rep}} - 1\}.$$

Committed rows. The committed showing witness consists of:

(1) the signature limb rows $L_{c,b,\ell}$;
(2) the carrier rows C^M and C^{ctr};
(3) the decoded non-sign source rows

$$M,\ K,\ R_0,\ R_1;$$

(4) the full replay-image block families

$$\widehat{T}_0, \ldots, \widehat{T}_{B_{\mathrm{rep}}-1}, \qquad \widehat{m}_0, \ldots, \widehat{m}_{B_{\mathrm{rep}}-1},$$

$$\widehat{R}_{0,0}, \ldots, \widehat{R}_{0,B_{\mathrm{rep}}-1}, \qquad \widehat{R}_{1,0}, \ldots, \widehat{R}_{1,B_{\mathrm{rep}}-1};$$

(5) the PRF companion rows.

The coefficient-domain signature witness u is reconstructed from the limb surface by the public digit-collapse map of the shortness gadget.

Parallel families. The compiled showing relation enforces:

(1) *Signature digit membership:* the limb rows satisfy the signed low-alphabet digit constraints of the chosen shortness profile.
(2) *Carrier membership:*

$$P_M(C^M) = 0, \qquad P_{\mathrm{ctr}}(C^{\mathrm{ctr}}) = 0.$$

(3) *Carrier-to-source decode:*

$$M = \mathrm{Dec}^{(1)}_M(C^M), \qquad K = \mathrm{Dec}^{(2)}_M(C^M),$$

$$R_0 = \mathrm{Dec}^{(1)}_{\mathrm{ctr}}(C^{\mathrm{ctr}}), \qquad R_1 = \mathrm{Dec}^{(2)}_{\mathrm{ctr}}(C^{\mathrm{ctr}}).$$

(4) *Packing:* the decoded rows M and K satisfy the same half-split message representation as in issuance.
(5) *Full replay residual:* for every block

$$b \in \{0, \ldots, B_{\mathrm{rep}} - 1\},$$

the replay aliases satisfy

$$(B^{\Theta}_{3,b} - \widehat{R}_{1,b}) \odot \widehat{T}_b - \left(B^{\Theta}_{0,b} + B^{\Theta}_{1,b} \odot \widehat{m}_b + B^{\Theta}_{2,b} \odot \widehat{R}_{0,b}\right) = 0.$$

(6) *PRF key binding and nonlinear constraints:* the companion rows are bound to the decoded K-coordinates, satisfy the checkpointed permutation equations, and bind the public tag after feed-forward and truncation.

Aggregated bridge families. The compiled showing relation includes the following aggregated families.

(1) *Signature-side bridges:* for every replay block b,

$$\widehat{T}_b = \mathcal{F}_{\mathrm{rep},b}(Au),$$

where u is the coefficient-domain signature witness reconstructed from the cert-side limb surface.
(2) *Non-sign transform bridges:* for every replay block b,

$$\widehat{m}_b = \mathcal{F}_{\mathrm{rep},b}(M + K), \qquad \widehat{R}_{0,b} = \mathcal{F}_{\mathrm{rep},b}(R_0), \qquad \widehat{R}_{1,b} = \mathcal{F}_{\mathrm{rep},b}(R_1).$$

Statement proved. The compiled showing proof asserts the existence of committed limb rows, source rows, replay-image rows, and PRF companion rows such that:

(1) the cert-side signature witness is short via the limb decomposition;
(2) the decoded source rows are the coefficient-domain message and randomness rows used throughout the statement;
(3) the full replay-image blocks are the exact transform images of Au, $M + K$, R_0, and R_1;
(4) the full replay residual vanishes blockwise; and
(5) the public tag is bound to the signed hidden key and the public nonce by the PRF companion constraints.

5.5 PRF constraints

The public tag is tied to the signed hidden key by a checkpointed arithmetization of the Poseidon2-like permutation from Section 4.3. The proof commits selected S-box outputs and lets the verifier replay all public linear wiring between checkpoints from openings and public constants. The resulting families are parallel PACS constraints over the companion rows, together with the public feed-forward and truncation equations that bind the final state to the tag.

5.6 Coefficient-domain consequence and statement strength

The completeness, knowledge-soundness, and zero-knowledge guarantees imported from SmallWood apply to the compiled PACS relations stated in this section. In particular, the showing proof directly certifies the full transform-basis replay relation on explicitly committed replay-image blocks together with the signature-side and non-sign bridge families. The coefficient-domain consequence is the following.

Lemma 5.1 (Exact replay transform). *The public replay map*

$$\mathcal{F}_{\mathrm{rep}} : R_q \longrightarrow \mathcal{T}$$

is the exact transform isomorphism used by the replay basis. In particular, it is additive, multiplicative, and injective.

Proof. The replay basis is chosen so that $\mathcal{F}_{\mathrm{rep}}$ is the corresponding CRT/NTT isomorphism of R_q. The stated properties are exactly the defining properties of that isomorphism. □

Lemma 5.2 (Coefficient-domain consequence of the compiled showing relation)**.** *Assume the compiled showing relation of Section 5.4. Then*

$$B_3Au - (Au)R_1 - B_0 - B_1(M+K) - B_2R_0 = 0$$

in R_q.

Proof. For every replay block b, the bridge families and the public lift definition give

$$\widehat{T}_b = \mathcal{F}_{\text{rep},b}(Au), \qquad \widehat{m}_b = \mathcal{F}_{\text{rep},b}(M+K),$$

$$\widehat{R}_{0,b} = \mathcal{F}_{\text{rep},b}(R_0), \qquad \widehat{R}_{1,b} = \mathcal{F}_{\text{rep},b}(R_1),$$

$$B^{\ominus}_{i,b} = \mathcal{F}_{\text{rep},b}(B_i) \qquad \text{for } i \in \{0,1,2,3\}.$$

Substituting these identities into the blockwise replay residual gives

$$\mathcal{F}_{\text{rep},b}(B_3Au) - \mathcal{F}_{\text{rep},b}((Au)R_1) - \mathcal{F}_{\text{rep},b}(B_0) - \mathcal{F}_{\text{rep},b}(B_1(M+K)) - \mathcal{F}_{\text{rep},b}(B_2R_0) = 0$$

for every b. Concatenating the blocks yields

$$\mathcal{F}_{\text{rep}}\big(B_3Au - (Au)R_1 - B_0 - B_1(M+K) - B_2R_0\big) = 0.$$

By Lemma 5.1, $\mathcal{F}_{\text{rep}}$ is injective. Hence

$$B_3Au - (Au)R_1 - B_0 - B_1(M+K) - B_2R_0 = 0$$

in R_q. □

Corollary 5.1 (Abstract rational-hash relation under admissibility). *If, in addition, the witness satisfies*

$$B_3 - R_1 \in R_q^{\times},$$

then

$$Au = h_{M+K,(R_0,R_1)}(B).$$

Proof. Lemma 5.2 gives

$$(B_3 - R_1) \odot Au - \big(B_0 + B_1(M+K) + B_2R_0\big) = 0.$$

Under the admissibility condition $B_3 - R_1 \in R_q^{\times}$, Lemma 2.8 implies

$$Au = h_{M+K,(R_0,R_1)}(B).$$

□

Corollary 5.2 (Consequence of an accepting post-sign proof). *Assume the post-sign NIZK is knowledge sound for the compiled showing relation. Then every accepting post-sign proof yields a witness (u, M, K, R_0, R_1) satisfying*

$$B_3Au - (Au)R_1 - B_0 - B_1(M+K) - B_2R_0 = 0$$

in R_q. If the extracted witness additionally satisfies denominator admissibility, then it also satisfies

$$Au = h_{M+K,(R_0,R_1)}(B).$$

Proof. Knowledge soundness yields a witness satisfying the compiled showing relation. Lemma 5.2 then gives the coefficient-domain cleared identity, and Corollary 5.1 gives the quotient-side relation under admissibility. □

Remark 5.3 (Statement strength). The theorem suite above proves a one-way consequence from the compiled full-replay showing relation to the coefficient-domain cleared identity, and from there to the abstract quotient relation under admissibility. It does not assert a converse implication from the coefficient-domain identity back to the full compiled relation. The present paper therefore does not claim a two-way equivalence between the compiled showing relation and the abstract showing relation.

Remark 5.4 (Strengthened post-sign variant). If one wishes the post-sign proof itself to certify the abstract quotient relation without any external admissibility side condition, the compiled statement can be strengthened by introducing an inverse witness z and enforcing

$$(B_3 - R_1) \odot z = 1, \qquad Au = z \odot \big(B_0 + B_1(M+K) + B_2R_0\big).$$

The present section does not assume that this stronger variant has been adopted.

The remaining sections use Lemma 5.2 and Corollary 5.2 whenever a coefficient-domain consequence of the compiled showing proof is needed, and invoke Corollary 5.1 only when admissibility is also available.

6 PARAMETERS AND INTEGRATION

This section explains how parameters interact across the ARC-SPRUCE stack: the lattice signature layer, the rational hash, the SmallWood proof system, and the PRF. Detailed parameter tables and extended benchmark data are deferred to Appendix D.

6.1 Parameter dependencies

All components share the same base ring R_q and modulus q:

- The signature map $A \in R_q^{n\times m}$ and hash key B are defined over R_q.
- SmallWood constraints are evaluated over the base field $\mathbb{F}_q$ and therefore use the same modulus q.
- The PRF is instantiated over $\mathbb{F}_q$ so that its constraints are native to the same field.

The coefficient bound $\beta_{\mathtt{msg}}$ governs the sampling ranges for the message and bounded randomness, the centering modulus $\Delta = 2\beta_{\mathtt{msg}} + 1$, and the membership polynomials used by the NIZKs.

6.2 Sampler parameters

The trapdoor sampler must output short signatures u with overwhelming probability and with a distribution close to a discrete Gaussian over the appropriate lattice coset. This requires choosing:

- ring degree N and modulus q,
- trapdoor quality,
- Gaussian widths for the sampler, and
- an acceptance predicate that yields a public verification bound $\beta_{\mathtt{sig}}$.

We use an NTRU-style trapdoor together with a Gaussian preimage sampler; details and tuning rules are given in Appendix B.

6.3 vSIS and rational-hash parameters

Rational-hash admissibility. The abstract quotient $h_{m,(r_0,r_1)}(B)$ is evaluated only when $B_3 - r_1 \in R_q^{\times}$. In the issuance model, this admissibility condition is enforced during honest target derivation by restart. The compiled pre-sign proof of Section 5.3 certifies only the cleared-denominator relation for the public target T.

No wrap-around in bounds. Membership checks for $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ are implemented via vanishing polynomials over $\mathbb{F}_q$, so one requires $\beta_{\mathtt{msg}} \ll q$ to avoid modular wrap-around artifacts.

Non-blind reduction parameters. The theorem-backed non-blind reduction imported from DKLW25 is parameterized by a norm bound β_{DKLW} in the norm used in [DKLW25a], by the inadmissibility bound δ, and by the predicate condition P. The public verifier bound $\beta_{\mathtt{sig}}$ used by ARC-SPRUCE should therefore not be identified with β_{DKLW} without a separate norm-comparison lemma. In this section, $\beta_{\mathtt{sig}}$ is recorded only as the public acceptance bound of the concrete verifier.

6.4 SmallWood parameters

SmallWood introduces parameters controlling packing, masking, and soundness:

- the packing size $s = |\Omega|$,
- the tail length ℓ for HVZK masking and degree enforcement,
- the spot-check count ℓ' and the number of masks ρ for PIOP soundness,
- the degree bounds for witness rows and constraint-derived masks, and
- the small-field knob θ when using an extension-field variant.

The pre-sign proof is dominated by carrier membership, decode bridges, linear replay on decoded aliases, and the transform-domain cleared relation for the public target. The post-sign proof adds the signature shortness gadget, the full replay-image showing relation, and the PRF checkpoint constraints.

Full replay-image geometry. Let $n_{\mathrm{cols}}^{\mathrm{rep}}$ denote the row width used for the replay blocks, and write

$$B_{\mathrm{rep}} := \left\lceil \frac{N}{n_{\mathrm{cols}}^{\mathrm{rep}}} \right\rceil.$$

In the full replay-image showing model, each replayed ring object is represented by B_{rep} replay blocks. Accordingly, the post-sign witness carries

$$4B_{\mathrm{rep}}$$

replay rows for the full images of Au, $M+K$, R_0, and R_1, in addition to the cert-side signature surface, source rows, and PRF companion rows.

Replay width as a proof-size knob. Increasing $n_{\mathrm{cols}}^{\mathrm{rep}}$ decreases the replay block count B_{rep} and hence the number of replay-image rows and associated bridge families. This tends to reduce transcript size. The tradeoff is that larger row width increases the degree of the committed row polynomials and therefore affects the LVCS/PCS soundness and opening costs. The sizing discussion should therefore treat $n_{\mathrm{cols}}^{\mathrm{rep}}$ as the main knob controlling the transcript-size tradeoff of the full replay-image showing proof.

Replay-transform obligations for the showing theorem. The coefficient-domain consequence theorem of Section 5.6 relies on the exact full replay transform $\mathcal{F}_{\mathrm{rep}}$ and on the exact bridge equalities for the replay-image rows. In the full replay-image model, injectivity is supplied by the transform itself rather than by an additional zero-detection assumption on a reduced replay map.

6.5 PRF parameters

The PRF must be secure at the target security level, efficiently representable in constraints, and compatible with the base field $\mathbb{F}_q$. We use a Poseidon2-like permutation with feed-forward and truncation as a concrete instantiation. Its parameters include the state width t, round counts (R_F, R_P), MDS matrices, round constants, S-box exponent d, and truncation length ℓ_{tag}. We require $\ell_{\mathrm{tag}} \geq \lceil \lambda / \log_2(q) \rceil$ so the tag contains λ bits of entropy. The algorithm is given in Section 4.3, while the checkpointed constraint shape used in the post-sign proof is stated in Section 5.5. Appendix E records extended notes.

6.6 Sizing discussion

Concrete proof sizes depend jointly on the sampler parameters, the SmallWood masking parameters, the replay-block width, and the PRF checkpoint schedule. In the full replay-image showing model, enlarging the replay-block width decreases the number of committed replay rows, while the degree and opening costs of each row increase. Appendix D records representative parameter tuples and measurement data; the present section records only the qualitative coupling between those choices.

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A ADDITIONAL DEFINITIONS

This appendix collects definitions used by the main text.

A.1 Centering map and carries

Let $\mathsf{BoundB} \in \mathbb{Z}_{>0}$ and $\Delta = 2\mathsf{BoundB} + 1$. Define the centering map $\mathsf{center} : [-2\mathsf{BoundB}, 2\mathsf{BoundB}] \to [-\mathsf{BoundB}, \mathsf{BoundB}]$ by

$$\mathsf{center}(x) = \begin{cases} x & \text{if } x \in [-\mathsf{BoundB}, \mathsf{BoundB}], \\ x + \Delta & \text{if } x \in [-2\mathsf{BoundB}, -\mathsf{BoundB} - 1], \\ x - \Delta & \text{if } x \in [\mathsf{BoundB} + 1, 2\mathsf{BoundB}]. \end{cases}$$

Applied coefficient-wise, this map allows unbiased combination of two bounded values when at least one is uniform.

Lemma A.1 (Centering preserves uniformity)**.** *Fix any* $a \in [-\mathsf{BoundB}, \mathsf{BoundB}]$. *If* $b \xleftarrow{\$} [-\mathsf{BoundB}, \mathsf{BoundB}]$ *then* $\mathsf{center}(a + b)$ *is uniform over* $[-\mathsf{BoundB}, \mathsf{BoundB}]$.

A.2 Vanishing polynomials for membership constraints

Membership constraints are expressed using vanishing polynomials over $\mathbb{F}_q$ with signed lifting.

Definition A.2 (Vanishing polynomial for $[-\mathsf{BoundB}, \mathsf{BoundB}]$)**.** Let $\mathsf{BoundB} \in \mathbb{Z}_{>0}$. Define

$$P_{\mathsf{BoundB}}(X) = \prod_{i=-\mathsf{BoundB}}^{\mathsf{BoundB}} \left(X - \langle i \rangle_q\right) \in \mathbb{F}_q[X].$$

For $x \in \mathbb{F}_q$, we have $x \in [-\mathsf{BoundB}, \mathsf{BoundB}]$ (via signed lifting) iff $P_{\mathsf{BoundB}}(x) = 0$ in $\mathbb{F}_q$.

Definition A.3 (Vanishing polynomial for $\{-1, 0, 1\}$)**.** Define $P_1(X) = (X + 1)X(X - 1) \in \mathbb{F}_q[X]$. Then $P_1(x) = 0$ iff $x \in \{-1, 0, 1\}$.

B GAUSSIAN PREIMAGE SAMPLING

This appendix details the Gaussian preimage sampler used by the issuer to sign a target $t \in R_q^n$ by producing a short preimage $u \in R^m$ such that $Au = t$ in R_q^n. Our instantiation follows the vSIS–BBS setting [DKLW25a] and relies on an NTRU trapdoor generated via Antrag together with a hybrid two-plane sampler [ENS+23].

B.1 The preimage sampling problem

Given a public linear map $A \in R_q^{n \times m}$ and a target $t \in R_q^n$, the signer samples

$$u \leftarrow \mathsf{SampPre}(\mathsf{td}_A, t) \qquad \text{such that} \qquad Au \equiv t \pmod q$$

and u is short (according to a public bound β_{sig}). Equivalently, u lies in the coset lattice

$$\Lambda_\perp^t(A) := \{\, x \in R^m : Ax \equiv t \ (\mathrm{mod}\ q) \,\}.$$

We require: (i) correctness ($Au = t$ and the norm test fails only with negligible probability), (ii) (approx.) Gaussianity of u over the coset (statistical hiding of the trapdoor), and (iii) a tight tail bound that supports efficient ZK range/norm checks in the post-sign proof.

B.2 NTRU trapdoor form and quality

We work with the rank-2 NTRU module over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ and its quotient R_q. An NTRU public key has the form $h = gf^{-1} \bmod q$ for secret $f, g \in R$. The corresponding module is

$$\mathsf{L}_{\mathsf{NTRU}}(h) = \{(u, v) \in R^2 : uh - v \equiv 0 \ (\mathrm{mod} q)\}.$$

A trapdoor basis of $\mathsf{L}_{\mathsf{NTRU}}(h)$ is

$$B_{f,g} = \begin{pmatrix} f & F \\ g & G \end{pmatrix}, \qquad \text{with} \qquad fG - gF = q,$$

whose columns are short and satisfy $g/f \equiv G/F \equiv h \pmod q$. Following [ENS+23], we measure trapdoor quality via per-slot energies under the canonical embeddings φ_i, requiring a uniform window around q. Concretely, for a quality parameter $\alpha > 1$ we target

$$\frac{q}{\alpha^2} \le |\varphi_i(f)|^2 + |\varphi_i(g)|^2 \le \alpha^2 q \qquad \text{for all slots } i.$$

Smaller α corresponds to tighter trapdoors and, for fixed N, q, to shorter signatures for the same security.

B.3 Sampler definition and tuning

Our sampler is a ring-based hybrid two-plane sampler in the evaluation domain. The sampler's main tuning parameters are:

- modulus q and ring degree N,
- trapdoor quality α,
- acceptance radius R_{acc} (controls rejections and ℓ_∞/ℓ_2 tails),
- per-slot widths $\sigma_k[i]$ (two planes $k \in \{1, 2\}$).

Per-slot calibration. Let $\widetilde{b}_k[i]$ denote the per-slot Gram–Schmidt lengths for plane k at slot i. A standard calibration equalizes acceptance probability across slots:

$$\sigma_k[i] = \sqrt{\frac{\alpha^2 q\, R_{\mathrm{acc}}^2}{\|\widetilde{b}_k[i]\|^2} - R_{\mathrm{acc}}^2} \qquad (k \in \{1, 2\}).$$

This prevents weak slots and yields consistent output variance.

Hiding threshold (smoothing). To achieve statistical hiding, we enforce a floor $\sigma_{\min}$ such that $\sigma_k[i] \geq \sigma_{\min}$ for all k, i, where $\sigma_{\min}$ exceeds the smoothing parameter of the relevant lattice cosets. In practice, we apply a global scale factor to raise all $\sigma_k[i]$ above the theoretical floor with slack.

B.4 Annulus sampling (Antrag key generation)

Antrag replaces trial-and-error trapdoor generation with direct annulus sampling in the Fourier domain [ENS+23]. For each conjugate pair (there are $N/2$ independent slots), the key generator samples $(|\varphi_i(f)|, |\varphi_i(g)|)$ uniformly in an annulus arc $A^+(r, R)$ with

$$\sqrt{q}/\alpha \;<\; r \;<\; R \;<\; \alpha\sqrt{q},$$

chooses random phases, inverse-embeds to the coefficient domain, and rounds to obtain $(f, g) \in R^2$. By selecting a "middle" annulus window (see [ENS+23]) one ensures that the rounded trapdoor satisfies the desired α-window with high probability and with modest repetition.

B.5 Hybrid two-plane sampler (details)

We sketch the two-plane hybrid sampler in the notation of [ENS+23]. Let td_A encode an NTRU trapdoor for A and let $t \in R_q^n$ be the target. Write $\langle\cdot\rangle_q$ for signed coefficient lift and define center_q as coefficient-wise centering into $(-\frac{q}{2}, \frac{q}{2}]$.

Algorithm 3 SampPre(td_A, t): two-plane hybrid sampler (sketch)

Lift t to coefficients and set $c \leftarrow \mathrm{center}_q(t) \in (-\frac{q}{2}, \frac{q}{2}]^n$.
Using td_A, compute per-slot widths $\sigma_k[i]$ (two planes) calibrated to trapdoor quality.
repeat
    Sample slot-wise Gaussian pairs on the two planes, reconstruct a coefficient-domain candidate u.
    Accept iff the norm predicate holds for u (e.g., ℓ_2 and/or ℓ_∞ bound).
until accepted
return u.

Public relations. Downstream verification and proofs use only the public relation $Au = t$ and shortness of u (bounded by $\beta_{\mathtt{sig}}$), checked by the holder on receipt and enforced by the issuer's rejection sampling. The post-sign NIZK then proves knowledge of such a short u without revealing it.

C SMALLWOOD PROOF-STACK DETAILS

This appendix records the proof-stack geometry for the compiled relations of Section 5. It specifies committed witness-row families, public lifts, PACS splits, and the PCS/LVCS/DECS opening structure used by the prover and verifier. It does not strengthen the statement layer beyond Section 5; in particular, coefficient-domain consequences are obtained exactly by the theorem suite stated there.

C.1 Domains and packing

Fix a base field $\mathbb{F}_q$, an evaluation set $\Omega \subset \mathbb{F}_q$ of size s, and a disjoint tail set $\Omega' \subset \mathbb{F}_q \setminus \Omega$ of size ℓ. A witness row is interpolated into a polynomial of degree $\leq s + \ell - 1$ whose values on Ω are the packed coordinates and whose values on Ω' are random masks. The prover commits to these explicit-domain row polynomials, and the verifier later replays the active constraint residuals from openings of the same polynomials.

C.2 DECS (Degree-Enforcing Commitment Scheme)

DECS commits to evaluations of witness polynomials and enforces that they arise from low-degree polynomials rather than arbitrary vectors. In our instantiation, commitments are implemented by a Merkle tree over an NTT-sized domain, while degree soundness is obtained by checking masked low-degree relations at verifier-chosen indices.

Algorithm 4 DECS.DeriveGamma(root, η, r, q) $\longrightarrow \Gamma$ (verifier and prover)

Initialize counter $ctr \leftarrow 0$; let $L \leftarrow \left\lfloor \frac{2^{64}}{q} \right\rfloor \cdot q$.
for $k = 0$ to $\eta - 1$ **do**
    for $j = 0$ to $r - 1$ **do**
        repeat
            $x \leftarrow$ SHA256(root $\|$ LE64(ctr)) interpreted as a 64-bit integer
            $ctr \leftarrow ctr + 1$
        until $x < L$
        $\Gamma_{k,j} \leftarrow x \bmod q$
return Γ

Algorithm 5 DECS.VerifyEval(root, Γ, R, open) (verifier)

Require: Domain size N, modulus q, mask count η, committed row count r.
for each opened index e **do**
    Verify Merkle authentication for e against root; reject on failure.
    for each mask row $k < \eta$ **do**
        Check $R_k^\star[e] \stackrel{?}{=} M_k(e) + \sum_{j=0}^{r-1} \Gamma_{k,j}\, P_j(e) \pmod q$.
return accept

C.3 LVCS (Linear Vector Commitments)

LVCS lifts DECS into a commitment to packed witness rows that supports succinct openings of $\mathbb{F}_q$-linear forms of those rows. SmallWood uses LVCS both as a batching layer and as the core of its hash-based PCS: the verifier requests openings of linear forms corresponding to replay residuals, and DECS ensures that the opened values come from low-degree committed row polynomials.

Algorithm 6 LVCS.CommitInitWithParams(ringQ, $\{r_j\}_{j<r}$, ℓ, params) $\to$ (root, ProverKey) (prover)

Require: Ring R_q of size N, packed rows $r_j \in \mathbb{F}_q^{n_{\text{cols}}}$, tail length $\ell > 0$.
Sample $\text{mask}_j \xleftarrow{\$} \mathbb{F}_q^{\ell}$ for each row j (HVZK tails).
Interpolate each row to $P_j(X) \leftarrow \text{interpolateRow}(\text{ringQ}, r_j, \text{mask}_j, n_{\text{cols}}, \ell)$.
Commit to $\{P_j\}$ via DECS and obtain Merkle root root.
Derive $\Gamma \leftarrow \text{DECS.DeriveGamma}(\text{root}, \eta, r, q)$ and cache values needed for openings.
return (root, ProverKey).

Algorithm 7 LVCS.CommitStep1(root) $\to \Gamma$ (verifier)

Store root and derive $\Gamma \leftarrow \text{DECS.DeriveGamma}(\text{root}, \eta, r, q)$.
return Γ.

Algorithm 8 LVCS.CommitStep2($\{R_k\}_{k<\eta}$) $\to$ accept/reject (verifier)

for each mask row R_k **do**
    Reject if $\deg(R_k) > d$ (degree guard before openings).
return accept.

Algorithm 9 LVCS.EvalInitMany(ringQ, ProverKey, $\{\text{Coeffs}_k\}_{k<m}$) $\to \{\bar{v}_k\}_{k<m}$ (prover)

Require: Linear forms $\text{Coeffs}_k \in \mathbb{F}_q^r$ for $k = 0, \ldots, m-1$.
Let $\ell \leftarrow |\text{mask}_j|$.
for $k = 0$ to $m-1$ **do**
    Initialize $\bar{v}_k \leftarrow 0 \in \mathbb{F}_q^{\ell}$.
    Set $\bar{v}_k \leftarrow \sum_{j=0}^{r-1} \text{Coeffs}_k[j] \cdot \text{mask}_j \in \mathbb{F}_q^{\ell}$.
return $\{\bar{v}_k\}_{k<m}$.

Algorithm 10 LVCS.ChooseE(ℓ, n_{cols}) $\to E$ (verifier)

Require: Ring size N; packed domain size $n_{\text{cols}} = |\Omega|$; masked slab indices $[n_{\text{cols}}, n_{\text{cols}} + \ell)$.
Sample $E \subset \{n_{\text{cols}}+\ell, \ldots, N-1\}$ uniformly without replacement with $|E| = \ell$.
return E.

Algorithm 11 LVCS.EvalFinish(ProverKey, E) $\to$ Opening (prover)

open $\leftarrow$ ProverKey.DecsProver.EvalOpen(E).
return Opening{DECSOpen = open}.

Algorithm 12 LVCS.EvalStep2($\{\bar{v}_k\}$, E, open, C, v_{targets}) (verifier)

Require: Coeffs $C \in \mathbb{F}_q^{m\times r}$, public prefixes $v_{\text{targets}} \in (\mathbb{F}_q^{n_{\text{cols}}})^m$.
Split open into mask openings for $[n_{\text{cols}}, n_{\text{cols}} + \ell)$ and tail openings at E.
Verify the DECS openings (degree soundness and Merkle authentication); reject on failure.
for each $k < m$ and each masked index $i < \ell$ **do**
    Check $\sum_{j=0}^{r-1} C[k,j] \cdot \text{maskOpen.Pvals}[i][j] \stackrel{?}{=} \bar{v}_k[i] \pmod q$.
Reconstruct each batched polynomial $Q_k(X)$ from its Ω-prefix $v_{\text{targets}}[k]$ and mask tail $\bar{v}_k$.
Check tail consistency of Q_k against the opened linear combinations at points in E.
return accept.

C.4 LVCS and PCS/PIOP

LVCS provides the linear opening queries needed by the polynomial oracle model, while DECS provides degree soundness for the committed rows. SmallWood combines the two into a transparent hash-based PCS used by the PACS and PIOP layers.

C.5 What DECS/LVCS/PCS bind in ARC-SPRUCE

At a high level, the SmallWood stack binds *all committed witness rows of one statement* under a single PCS root, and it enforces that every opened value is consistent with one low-degree polynomial view of those rows. In our ARC instantiation this binding matters in three ways:

- **Row binding and low degree (DECS).** DECS ensures that every committed row is the low-degree polynomial matching the packed values on Ω and the random tails on Ω'.
- **Replay binding across subsystems (LVCS).** The verifier reuses the same opened row values to replay every active constraint family. In issuance this covers carrier membership, decode bridges, commitment, centering, packing, the transform-domain public-target hash, and the four non-sign transform bridges. In showing it covers signature shortness, the signature-side bridges to the full replay-image family $\widehat{T}_0, \ldots, \widehat{T}_{B_{\text{rep}}-1}$, the non-sign bridges to $\widehat{m}_0, \ldots, \widehat{m}_{B_{\text{rep}}-1}$, $\widehat{R}_{0,0}, \ldots, \widehat{R}_{0,B_{\text{rep}}-1}$, and $\widehat{R}_{1,0}, \ldots, \widehat{R}_{1,B_{\text{rep}}-1}$, the blockwise replay residuals, and the PRF companion constraints.
- **Single committed witness.** Because all committed rows live under one PCS root, the prover cannot use one hidden message for the hash relation and another for the PRF or the commitment. All replay checks refer to one coherent witness assignment.

C.6 Openings requested in issuance and showing

Both proofs follow the same pattern: the prover commits to all statement rows as explicit-domain polynomials, the verifier derives Fiat–Shamir challenges, and the prover opens the committed rows at verifier-chosen points. The verifier then recomputes the active residuals from those openings and checks the parallel replay identities at E' together with the aggregated Ω-sum identities.

Pre-sign witness surface and replay. The committed pre-sign rows are

$$C^M,\ C^{\mathrm{preRU}},\ C^{\mathrm{preR}},\ C^{\mathrm{ctr}},\ C^J,$$
$$M,\ K,\ R_0^U,\ R_1^U,\ R,\ R_0,\ R_1,\ J_0,\ J_1,$$
$$\widehat{M},\ \widehat{K},\ \widehat{R}_0,\ \widehat{R}_1.$$

The verifier derives the public lifts $B_0^\Theta, B_1^\Theta, B_2^\Theta, B_3^\Theta, T^\Theta$ and replays the carrier membership relations, decode bridges, commitment, centering, packing, the four transform bridges, and the transform-domain cleared relation

$$(B_3^\Theta - \widehat{R}_1) \odot T^\Theta - \big(B_0^\Theta + B_1^\Theta \odot (\widehat{M} + \widehat{K}) + B_2^\Theta \odot \widehat{R}_0\big) = 0.$$

This appendix therefore records the compiled relation $R_{\mathrm{Issue}}^{\mathrm{cmp}}$ only.

Post-sign witness surface and replay. The committed post-sign rows consist of:

(1) the signature limb rows;
(2) the carrier rows C^M and C^{ctr};
(3) the decoded non-sign source rows M, K, R_0, R_1;
(4) the full replay-image block families
$$\widehat{T}_0, \ldots, \widehat{T}_{B_{\mathrm{rep}}-1}, \qquad \widehat{m}_0, \ldots, \widehat{m}_{B_{\mathrm{rep}}-1},$$
$$\widehat{R}_{0,0}, \ldots, \widehat{R}_{0,B_{\mathrm{rep}}-1}, \qquad \widehat{R}_{1,0}, \ldots, \widehat{R}_{1,B_{\mathrm{rep}}-1};$$
(5) the PRF companion rows.

The verifier replays:

(1) the shortness families on the cert-side signature surface;
(2) the carrier membership and decode families;
(3) the signature-side bridges
$$\widehat{T}_b = \mathcal{F}_{\mathrm{rep},b}(Au)$$
for every block b;
(4) the non-sign transform bridges
$$\widehat{m}_b = \mathcal{F}_{\mathrm{rep},b}(M+K), \qquad \widehat{R}_{0,b} = \mathcal{F}_{\mathrm{rep},b}(R_0), \qquad \widehat{R}_{1,b} = \mathcal{F}_{\mathrm{rep},b}(R_1)$$
for every block b;
(5) the blockwise replay residual
$$(B_{3,b}^\Theta - \widehat{R}_{1,b}) \odot \widehat{T}_b - \big(B_{0,b}^\Theta + B_{1,b}^\Theta \odot \widehat{m}_b + B_{2,b}^\Theta \odot \widehat{R}_{0,b}\big) = 0$$
for every block b; and
(6) the PRF checkpoint constraints.

The theorem stated in Section 5.6 then transports this full replay-image statement to the coefficient-domain cleared relation.

Replay-transform realization. The replay transform

$$\mathcal{F}_{\mathrm{rep}} : R_q \to \mathcal{T}$$

is the exact public transform used by the replay basis, partitioned into the replay blocks

$$\mathcal{F}_{\mathrm{rep},0}, \ldots, \mathcal{F}_{\mathrm{rep},B_{\mathrm{rep}}-1}.$$

The bridge families above certify that the replay-image rows are exactly the block projections of $\mathcal{F}_{\mathrm{rep}}(Au)$, $\mathcal{F}_{\mathrm{rep}}(M+K)$, $\mathcal{F}_{\mathrm{rep}}(R_0)$, and $\mathcal{F}_{\mathrm{rep}}(R_1)$. These are precisely the identities used in Lemma 5.2.

The Ω-sum check. The Ω-sum check is active in both proofs. In issuance it enforces the four non-sign transform bridges. In showing it enforces the full signature-side bridge family together with the full non-sign transform bridges. Thus the Σ_Ω check is part of the statement layer in both issuance and showing.

C.7 PACS/PIOP interface

SmallWood's PACS/PIOP layer checks that constraint polynomials vanish on Ω and at random points outside Ω, by committing to witness rows and mask polynomials and then opening them at verifier-chosen coordinates. The following algorithms summarize the high-level structure used in our proofs.

Algorithm 13 PACS.PIOPCommitOracle$_{\mathrm{Prover}}$(params) $\to$ pcsCom

1: Interpolate witness rows into $P_1, \ldots, P_n \in \mathbb{F}_q[X]_{\le s+\ell'-1}$ with ℓ' random blind evaluations outside Ω.
2: Sample mask polynomials $M_1, \ldots, M_\rho \in \mathbb{F}_q[X]_{\le d_Q}$ such that $\sum_{\omega\in\Omega} M_i(\omega) = 0$ for all i.
3: Commit to $P\|M$ via PCS (implemented through LVCS/DECS); obtain pcsCom.
4: **return** pcsCom.

Algorithm 14 PACS.PIOPBatchCombine$_{\mathrm{Prover}}$(pcsCom, Γ') $\to Q$

1: Using P, M, form batched polynomials
$$Q_i(X) \ = \ M_i(X) + \sum_j \Gamma'_{i,j}(X)\, F_j(X) + \sum_u \gamma'_{i,u}\, F'_u(X),$$
where F_j and F'_u are the parallel and aggregated constraint polynomials evaluated on P.
2: **return** $Q = (Q_1, \ldots, Q_\rho)$.

Algorithm 15 PACS.PIOPOpenAndCheck(pcsCom, Q) (verifier)

1: Sample a query set $E' \subset S$ with $|E'| = \ell'$ (Fiat–Shamir in NIZK).
2: Verify PCS openings of P and M at E' (via LVCS/DECS); reject on failure.
3: Check that each $Q_i(e)$ matches its definition from $M_i(e)$ and the constraint evaluations at $e \in E'$.
4: Check the masked Ω-sum test $\sum_{\omega\in\Omega} Q_i(\omega) = 0$ for all i.
5: **return** accept/reject.

In explicit-domain mode the prover also supplies the formal coefficient descriptions of the active parallel and aggregated families. This ensures that decode compositions, theta-lifted replay rows, and aggregated bridge polynomials are replayed as polynomial identities at E'.

C.8 Fiat–Shamir compilation

SmallWood compiles the interactive PACS/PIOP protocol to a NIZK using Fiat–Shamir with grinding, following [FR25b]. In our instantiation both issuance and showing use non-empty parallel and aggregated families, so the verifier derives one batched challenge schedule and checks both the replay identities at E' and the active Ω-sum constraints.

Algorithm 16 PACS.ToNIZK (Fiat–Shamir wrapper with grinding)

1: Sample salt $\xleftarrow{\$} \{0,1\}^{2\lambda}$.
2: Derive challenges by hashing the transcript with grinding counters to obtain required prefix properties.
3: Output PCS/LVCS/DECS openings and any required batched values; the verifier recomputes challenges and checks.

D EXTENDED PARAMETERS AND BENCHMARKS

This appendix provides extended parameter discussion and additional benchmark data.

D.1 Admissibility checklist

A parameter tuple is *admissible* for ARC-SPRUCE if it simultaneously satisfies:

(1) **Bound vs. modulus:** $\beta_{\mathtt{msg}} \ll q$ so membership checks on $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ do not wrap modulo q.
(2) **Sampler hiding and shortness:** Gaussian widths exceed the required smoothing threshold and the signature bound $\beta_{\mathtt{sig}}$ holds with overwhelming probability.
(3) **Rational-hash guard:** denominator invertibility succeeds with negligible failure probability under the holder randomness distribution.
(4) **SmallWood degree/soundness:** the DECS/LVCS degree bounds and PIOP soundness parameters meet the target security λ.
(5) **PRF entropy:** truncation length ℓ_{tag} yields at least λ bits of tag entropy.

D.2 Smoothing and sampler budgets

Let $\eta_\varepsilon(\Lambda)$ denote the smoothing parameter of a lattice Λ. A sufficient condition for statistical hiding in Gaussian preimage sampling is to choose Gaussian widths above a floor $\sigma_{\min} \geq \eta_\varepsilon(\Lambda_\perp(A))$, with slack (Appendix B). For the ambient lattice $\mathbb{Z}^{2N}$ one may use the classical bound

$$\eta_\varepsilon(\mathbb{Z}^{2N}) \;=\; \sqrt{\ln\!\big(4N(1+2^\lambda)\big)/\pi}.$$

In our tuning, trapdoor quality α and acceptance parameters are chosen so that the output shortness bound $\beta_{\mathtt{sig}}$ holds with overwhelming probability.

D.3 Example lattice parameters

Table 1 records one concrete instantiation compatible with the sizing discussion in Section 6.

Knob / Result	Symbol	Value (example)
Ring degree	N	1024
Modulus	q	1038337
Security target	λ	128
Trapdoor quality	α	$\approx$ 1.23 (measured)
Gaussian width scale	σ	tuned per-slot (Appendix B)
Signature bound	$\beta_{\mathtt{sig}}$	derived from sampler tail bound

Table 1: Example lattice parameters (illustrative; see Appendix B for the sampling interface and tuning).

D.4 SmallWood soundness terms

SmallWood's soundness is a combination of degree soundness (DECS), LVCS binding, and PIOP sampling errors [FR25b]. For parameters $(N, s, \ell, \ell', \rho, \eta)$ and degree bounds $d_{\text{decs}} = s + \ell - 1$ and d_Q, typical upper bounds take the form:

$$\varepsilon_{\text{decs}} \;\lesssim\; \binom{N}{d_{\text{decs}}+2} q^{-\eta}, \qquad \varepsilon_{\text{sum}} = q^{-\rho}, \qquad \varepsilon_{\text{tail}} \approx \frac{\binom{d_Q}{\ell'}}{\binom{|S|}{\ell'}}.$$

Our issuance statement uses only linear constraints (given public T), so d_Q is driven by membership checks; the showing statement additionally depends on the PRF effective degree (Appendix E).

D.5 SmallWood knobs (example)

For the issuance and showing proofs discussed in Section 6, one example parameter tuple is:

$$(s, \ell, \ell', \rho, \theta, \eta) \;=\; (16, 25, 2, 2, 4, 19),$$

with masking degree d_Q derived from the instantiated constraint degrees. The pre-sign statement remains linear in the witness (given public T), while the post-sign statement's effective degree is dominated by the PRF S-box composition.

D.6 Extended benchmark grid (SmallWood)

For additional context on how SmallWood knobs affect proof size and proving time, Table 2 reports an example sweep over packing and soundness parameters (small-field variant).

ProofKB	Time (s)	Bits	NCols	ℓ	ℓ'	ρ	θ	η
9.32	0.266	133.44	6	22	2	2	4	17
10.07	0.228	146.93	4	24	2	2	4	17
11.52	0.250	133.22	4	28	2	2	4	17
14.06	0.630	198.01	4	34	8	5	2	26
14.17	0.654	192.26	6	34	8	5	2	26
14.24	0.670	198.48	4	34	8	6	2	26

Table 2: Example sweep over SmallWood knobs (proof size, prover time, and soundness bits). Sizes exclude public parameters.

E PRF DETAILS AND MISCELLANEOUS

This appendix records a concrete ZK-friendly PRF instantiation compatible with the canonical ARC protocol.

E.1 PRF specification

We use a Poseidon2-like permutation with feed-forward and truncation. Let $k \in \mathbb{F}_q^{\ell_{\text{key}}}$ be the secret key (derived from the signed $\mathbf{k}$) and $n \in \mathbb{F}_q^{\ell_{\text{nonce}}}$ encode the public nonce. Let $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ be the permutation width. Define

$$\mathsf{F}(k, n) \;=\; \mathrm{Tr}\big(P(k\|n) + (k\|n)\big),$$

where $\mathrm{Tr} : \mathbb{F}_q^t \to \mathbb{F}_q^{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes and P is a Poseidon2-style permutation.

E.2 Permutation structure

Let $d \geq 3$ be the smallest constant such that $\gcd(d, q-1) = 1$. Fix round counts R_F (external rounds, even) and R_P (internal rounds), round constants, and MDS matrices M_E, M_I. The permutation has the form

$$P \;=\; E_{R_F} \circ \cdots \circ E_{R_F/2+1} \circ I_{R_P} \circ \cdots \circ I_1 \circ E_{R_F/2} \circ \cdots \circ E_1.$$

Each external round applies the S-box to all lanes, while each internal round applies the S-box to lane 1 only. This structure reduces nonlinear constraints in the post-sign NIZK while retaining diffusion.

External and internal rounds. Let $x = (x_1, \ldots, x_t) \in \mathbb{F}_q^t$. For each external round $i \in [1, R_F]$, fix per-lane constants $(c_1^{(i)}, \ldots, c_t^{(i)}) \in \mathbb{F}_q^t$. For each internal round $i \in [1, R_P]$, fix a scalar constant $\hat{c}^{(i)} \in \mathbb{F}_q$. Then:

$$\begin{aligned} E_i(x) \;&=\; M_E \cdot \big((x_1 + c_1^{(i)})^d, \ldots, (x_t + c_t^{(i)})^d\big)^\top, \\ I_i(x) \;&=\; M_I \cdot \big((x_1 + \hat{c}^{(i)})^d,\ x_2, \ldots, x_t\big)^\top. \end{aligned}$$

The matrices $M_E, M_I \in \mathbb{F}_q^{t\times t}$ are chosen so that they are invertible and provide sufficient diffusion (e.g., Cauchy-style MDS matrices as in Poseidon2).

E.3 Parameter generation

All PRF parameters are public. A typical parameter generator fixes:

- the base field modulus q (shared with the rest of the protocol),
- the S-box exponent d with $\gcd(d, q-1) = 1$,
- width $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ and truncation ℓ_{tag},
- round counts R_F, R_P,
- MDS matrices M_E, M_I and round constants $\{c^{(i)}\}, \{\hat{c}^{(i)}\}$.

The truncation length is chosen to satisfy $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$, ensuring that the tag contains λ bits of entropy.

E.4 Arithmetization strategy (System A with checkpoints)

To keep showing proofs compact, we commit only selected nonlinear checkpoints:

- **Key lanes.** We commit one row per key lane $\mathsf{Key}[i]$ and enforce equality to an encoding of signed $\mathbf{k}$ using selector constraints at fixed Ω-coordinates.
- **S-box outputs.** We commit only the S-box outputs at a checkpoint schedule (e.g., every internal round and periodically among external rounds). Linear wiring between checkpoints (add-round-constants, MDS multiplications) is recomputed by the verifier from opened evaluations and public constants.

This yields a System-A style constraint system where the dominant degree comes from composing at most a small number of unchecked S-box layers between checkpoints.